Region-based Deep Learning for Point Spread Function Localization

BSc Thesis Defense

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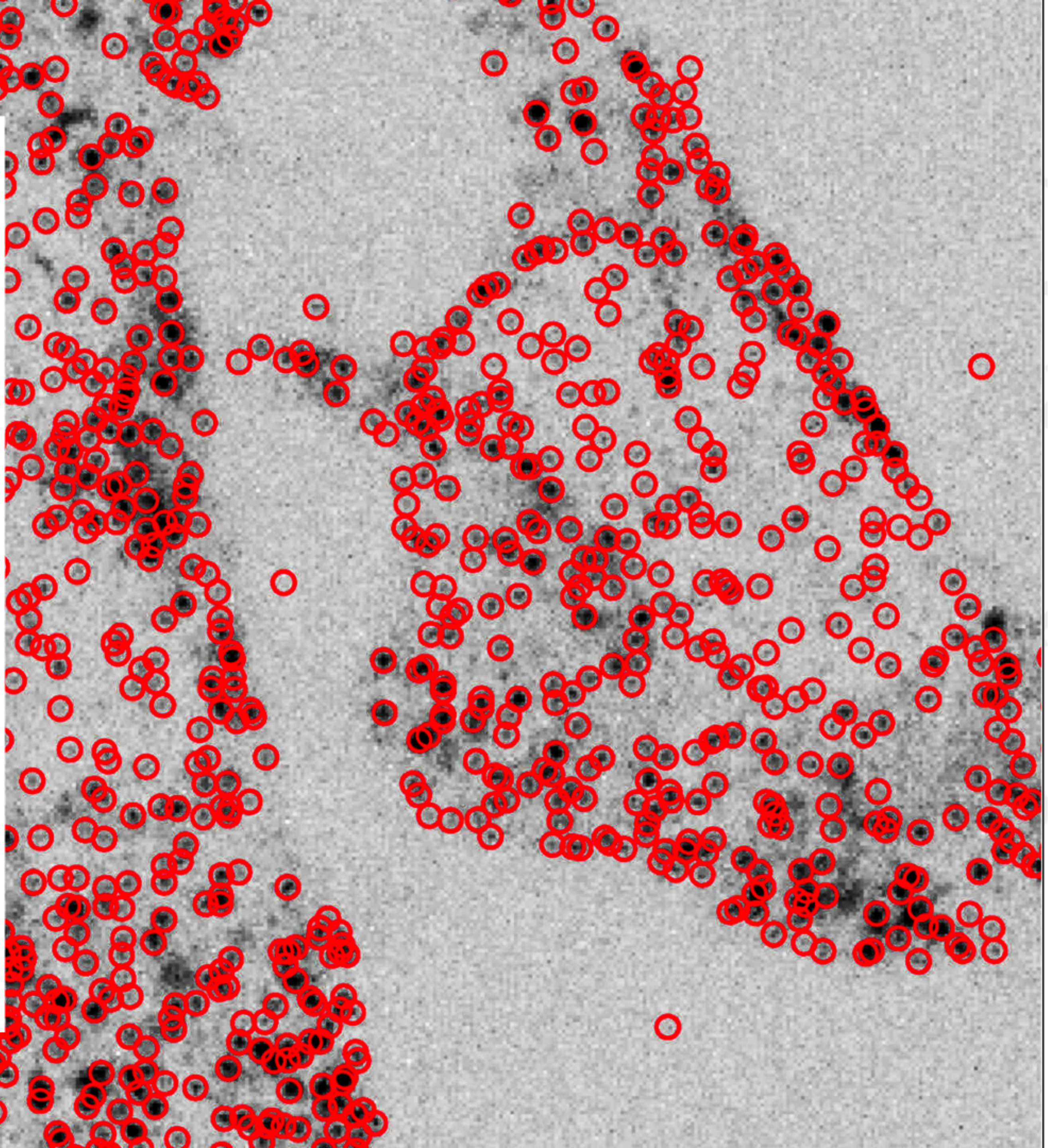
Department of Chemistry
June 20th 2025
KØBENHAVNS UNIVERSITET

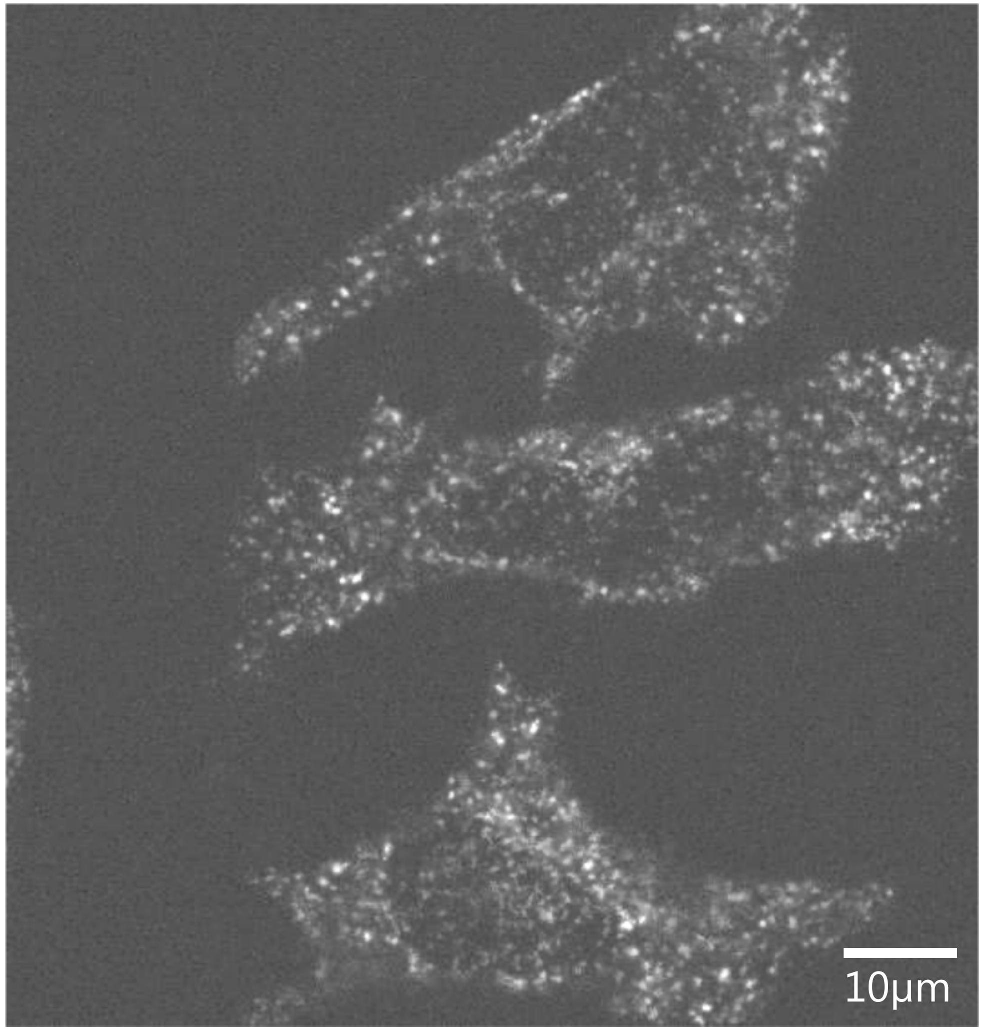


Presentation outline

- Introduction
- Aim of the project
- Methods: Data simulation and model architecture
- Results: Simulated and experimental data
- Conclusion

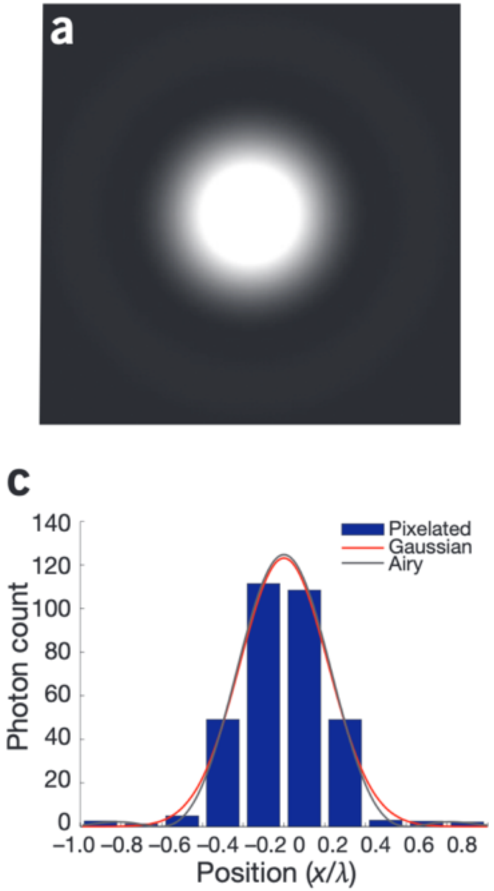
Fluorescence Microscopy is an essential tool in single-molecule studies

Collected by Sara Bleshøy

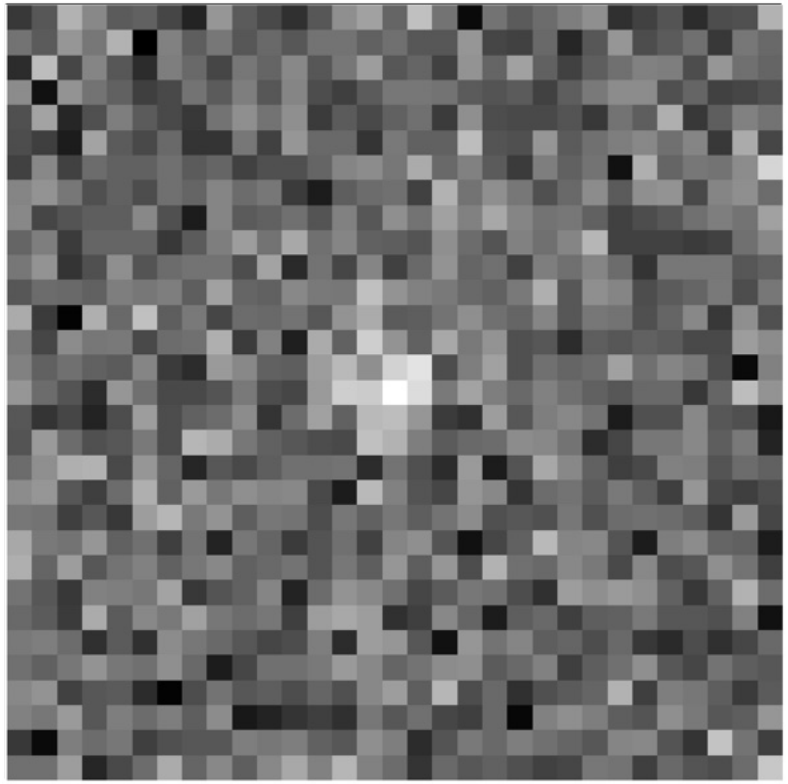


Transferrin proteins in HeLa cells.

Single-molecule studies require efficient point spread function (PSF) localization

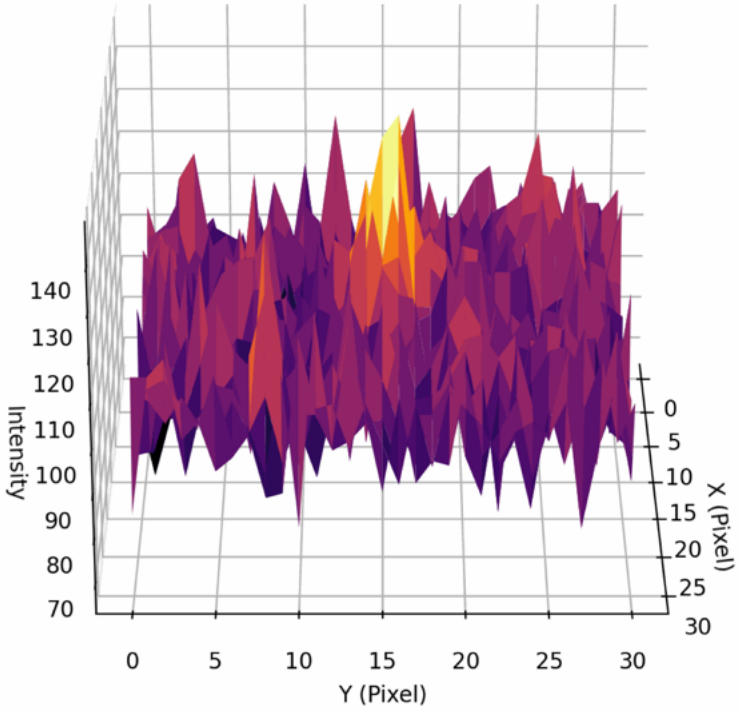


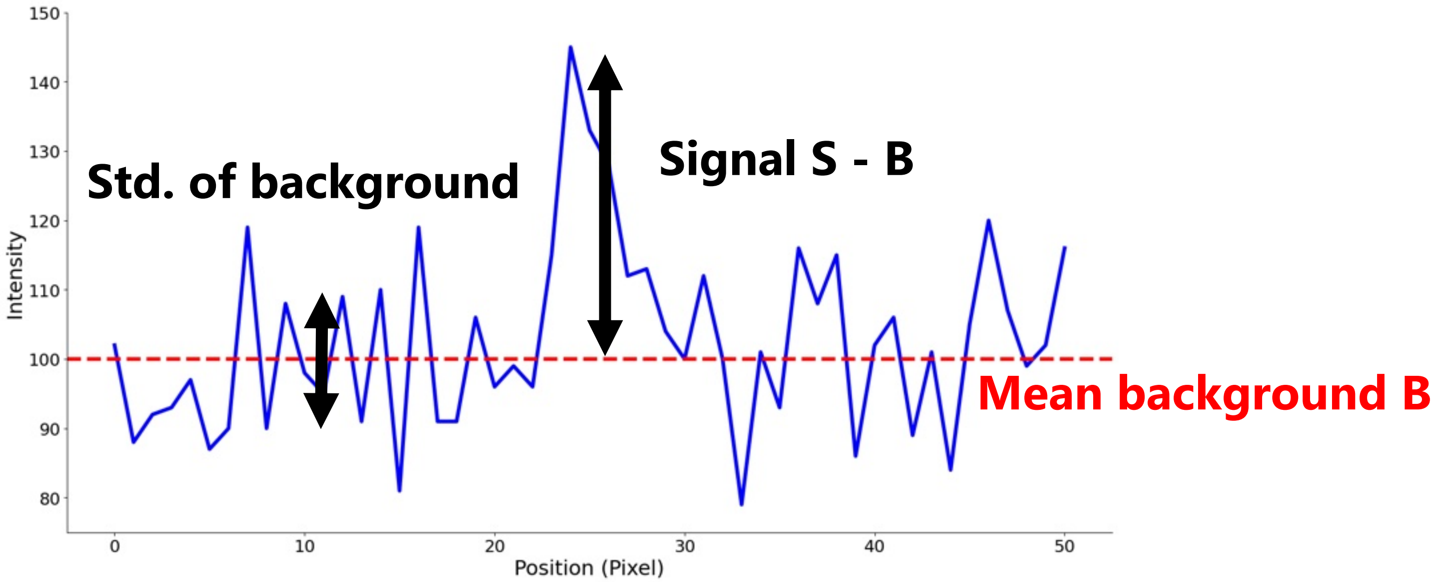
Small and Stahlheber, 2014



Shot-noise (Poisson distributed) makes localization difficult

Low signal-to-noise ratio (SNR) complicates localization





Assuming Poisson distribution,

$$\lambda = E(X) = Var(X)$$

$$\Rightarrow \sqrt{\mu} = \sigma$$

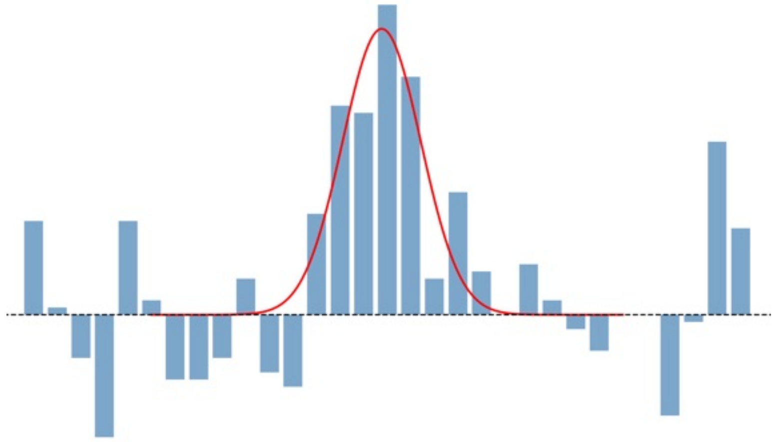
$$\mathrm{SNR_{bg}} = \frac{S - B}{\sqrt{B}}$$

$$\mathrm{SNR_{sig}} = \frac{S - B}{\sqrt{S}}$$

Current methods for PSF localization

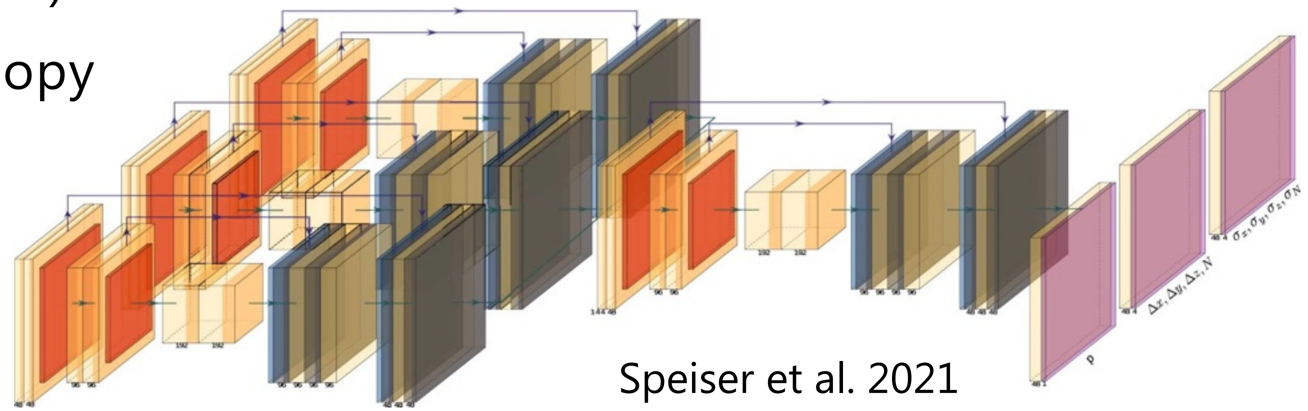
Conventional methods

- Ie. Gaussian fitting to data:
- Maximum Likelihood Estimation
- Least Squares Fitting



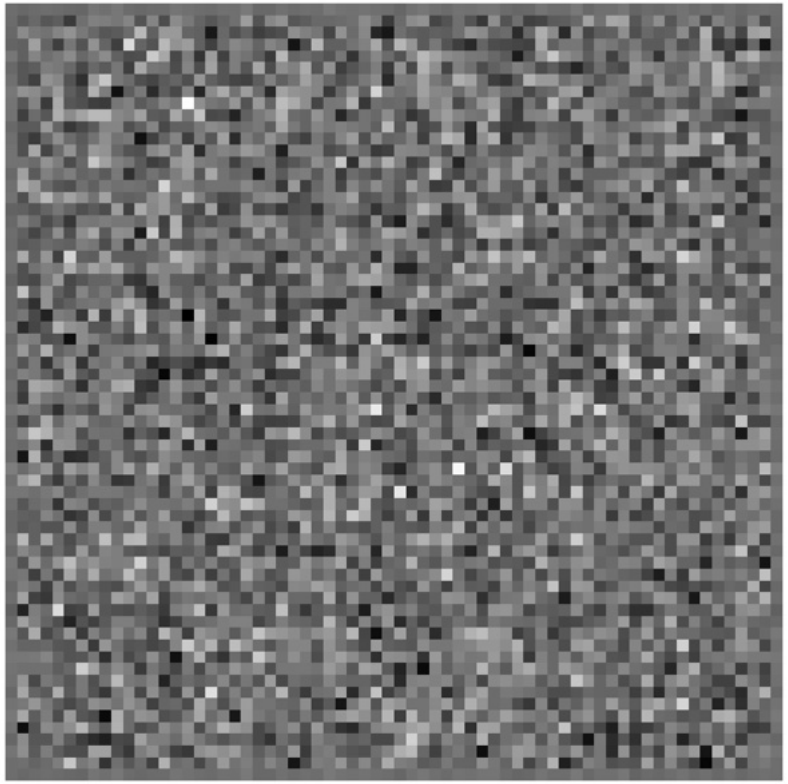
Deep Learning methods

- Uses convolutional neural networks (CNNs)
- For Single Molecule Localization Microscopy
- Eg. DECODE, DeepStorm3D

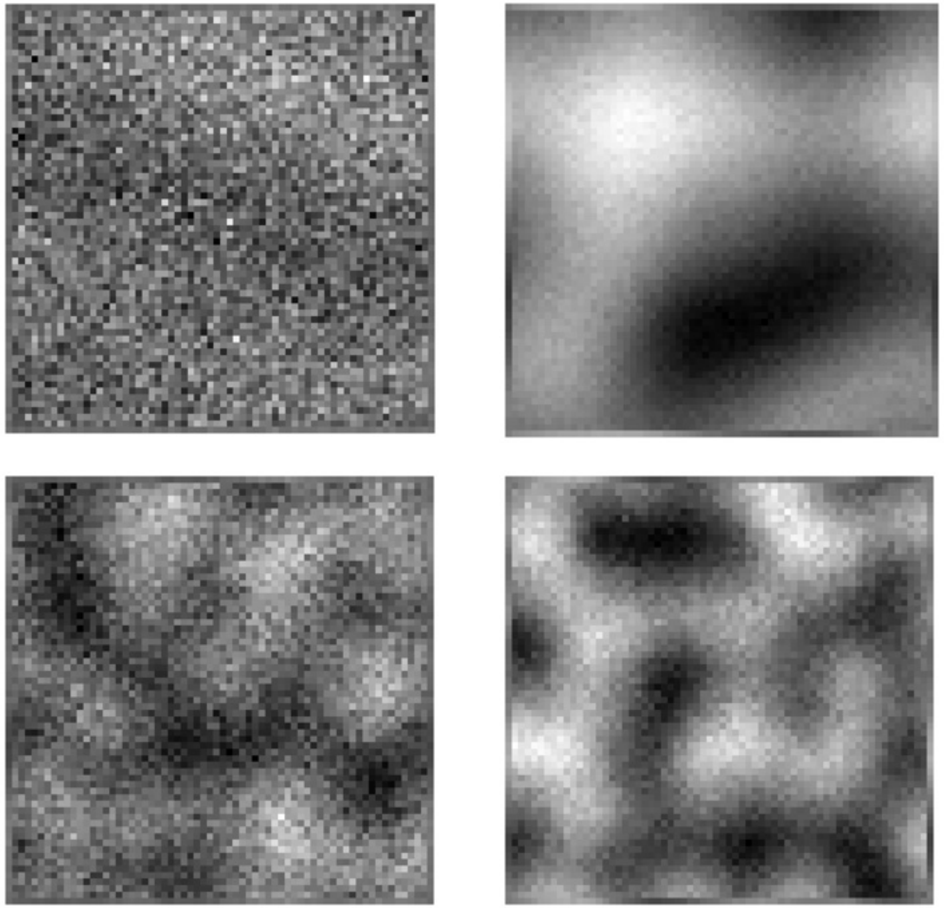


Speiser et al. 2021

Local image variability challenges accurate PSF localization



Homogeneous background



Heterogeneous background

Aim of the project

- Current methods for PSF localization are limited and not robust in heterogeneous environments

The aim of this project

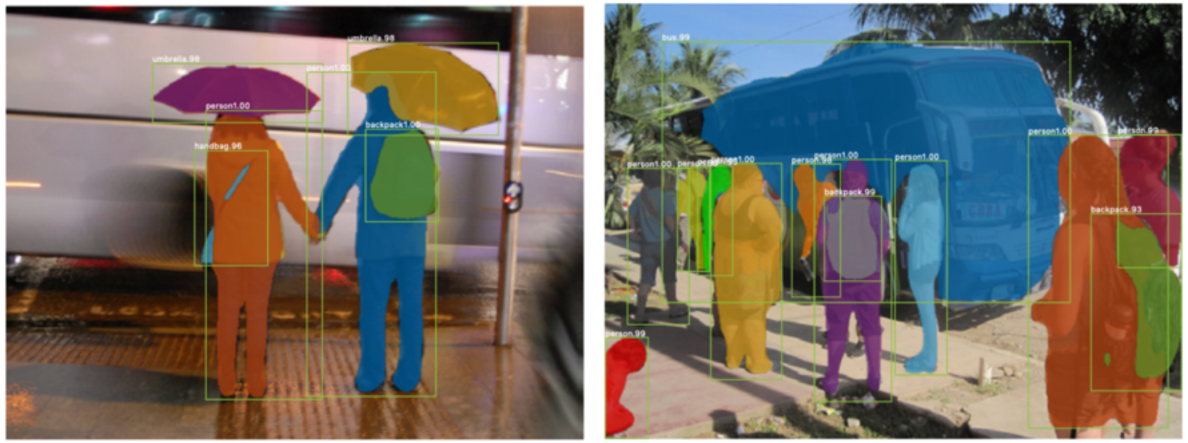
- Implement DL for PSF Localization in heterogeneous background
- Detection and classification of non-PSF objects
- Simulate data for efficient training

- Along the way… Learn Machine Learning, PyTorch, Git, Object Oriented Programming etc.

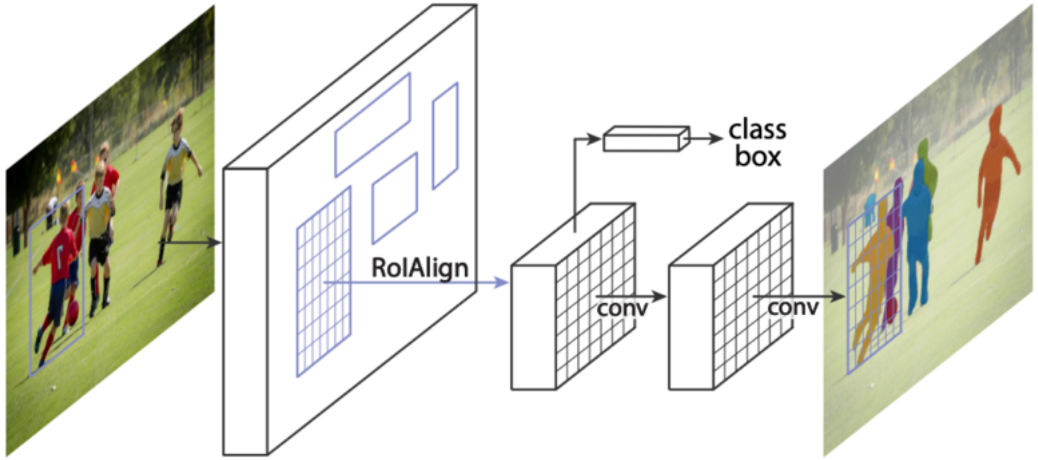
Choosing the model architecture: Mask Regional-CNN

Mask R-CNN is a model that does:

- Object instance detection
- Classification
- Binary segmentation

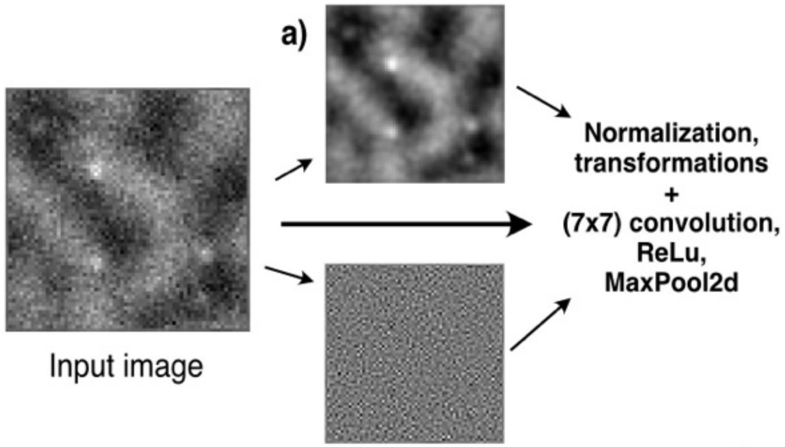


Mask R-CNN extends Faster R-CNN



He et al. 2017

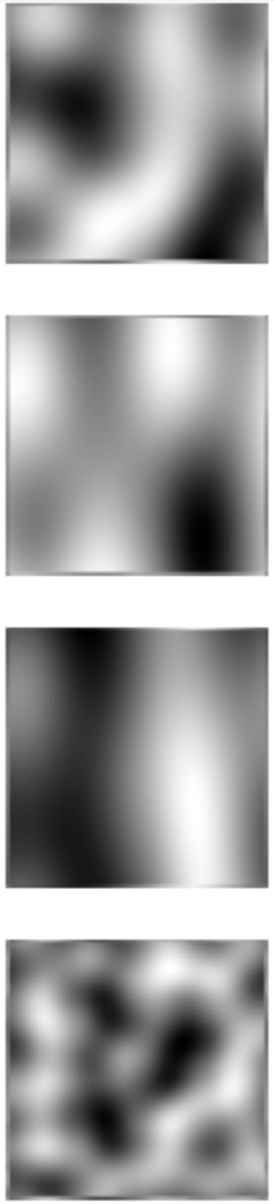
An in-depth look at model architecture



Data simulation

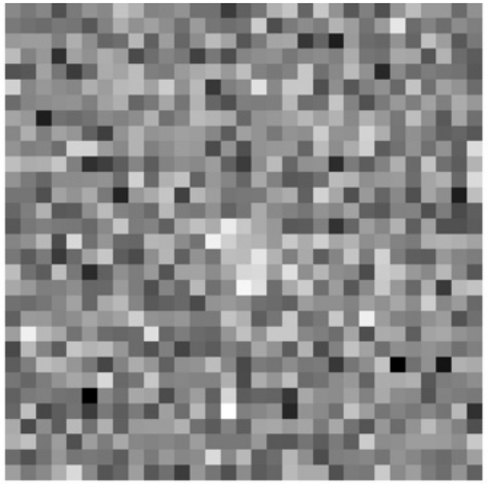
1. Perlin Noise for heterogeneous backgrounds
2. Draw perfect PSFs
3. Add shot-noise

Background

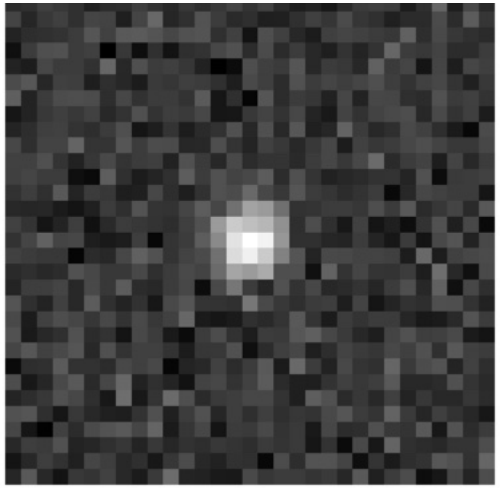


Training the model

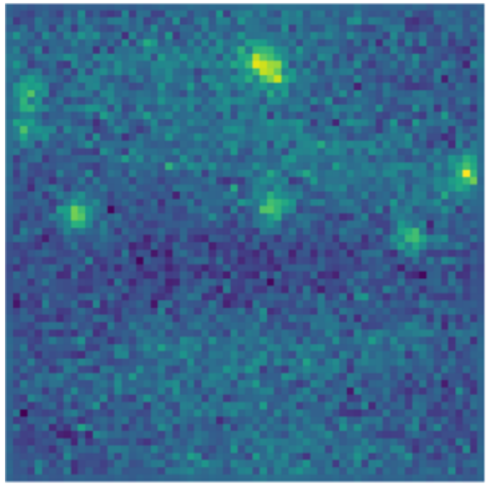
- Data simulation is fast and on-demand
- Trained on >600,000 unique images
- Backbone was trained from scratch (no pretrained weights)
- Large range of parameters (SNR, sizes, density, heterogeneity)



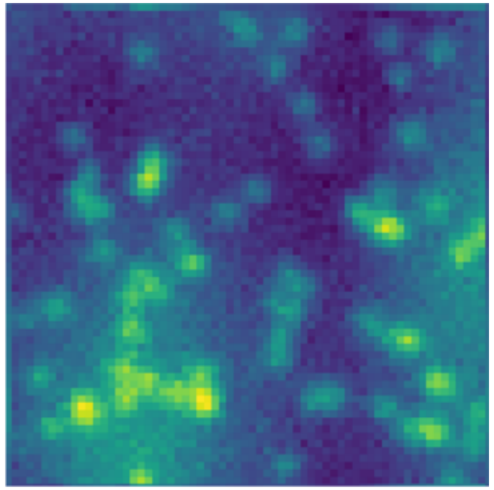
Low SNR ~ 3



High SNR ~9

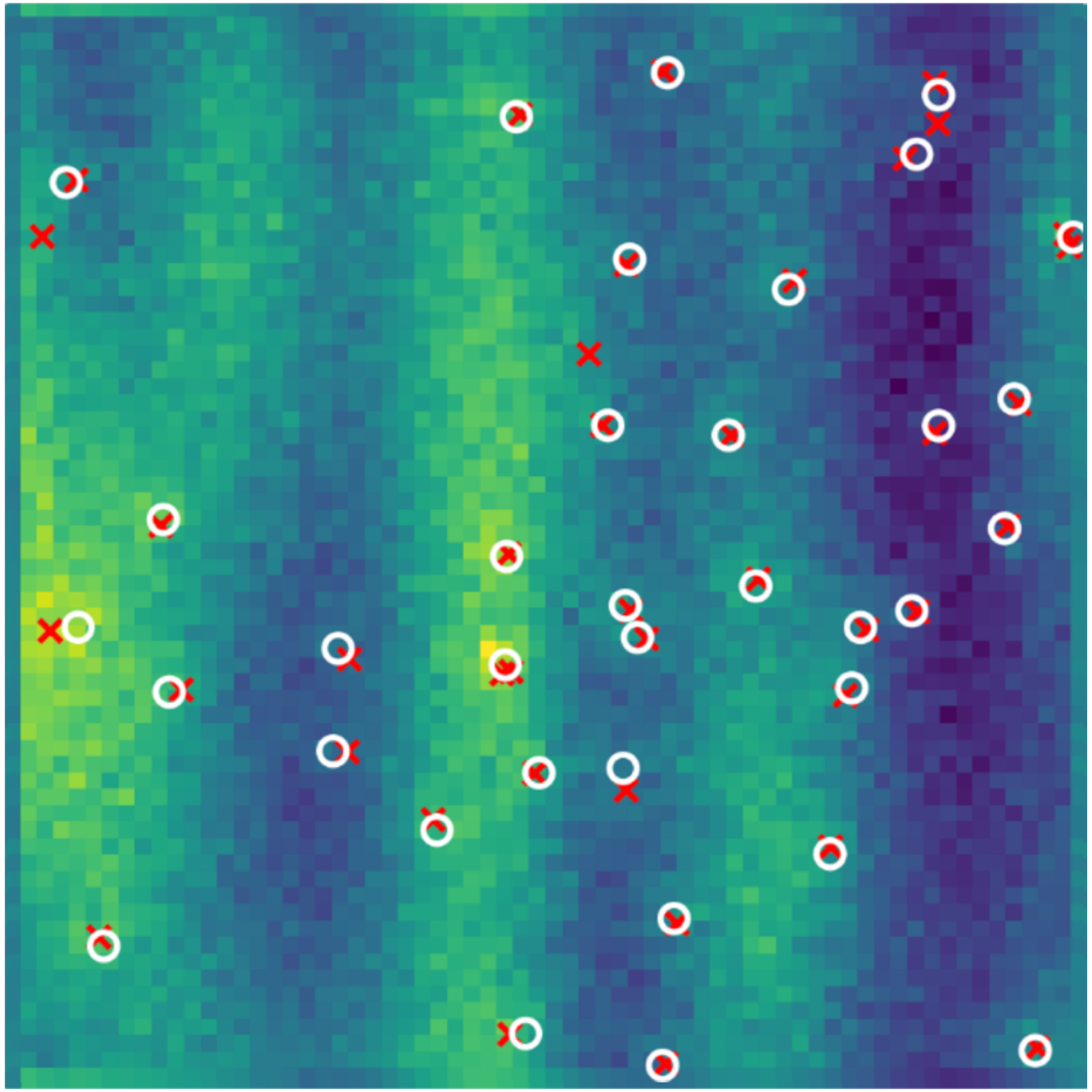


Low density

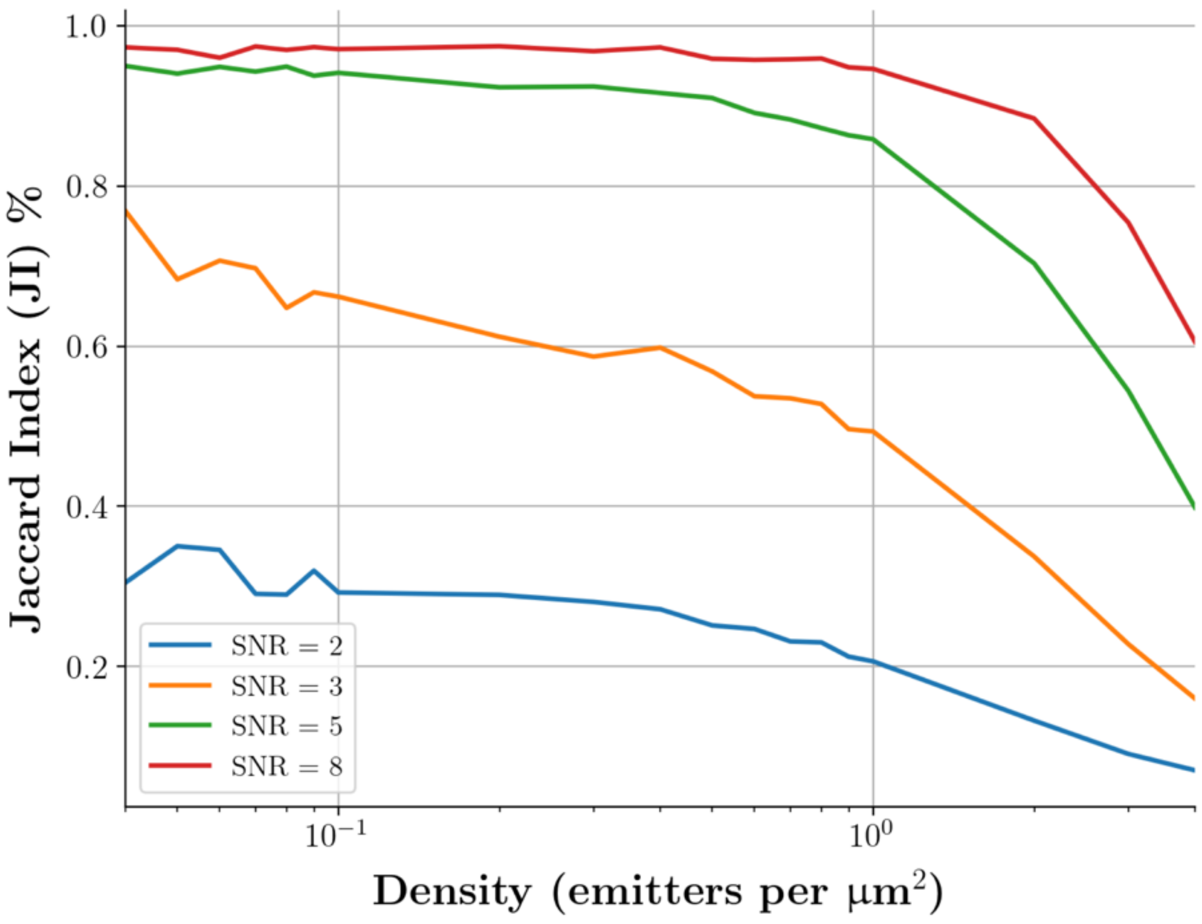


High density

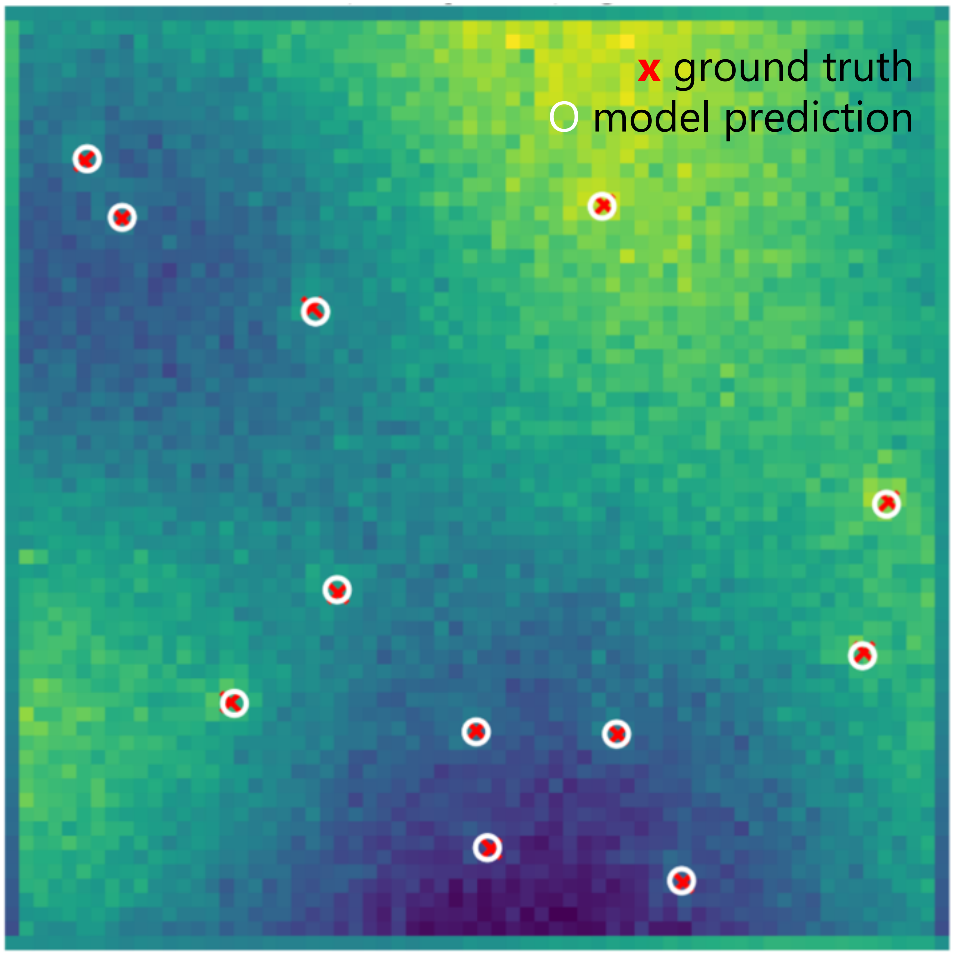
The results are in!



Detection accuracy 80-90% for medium SNR and density

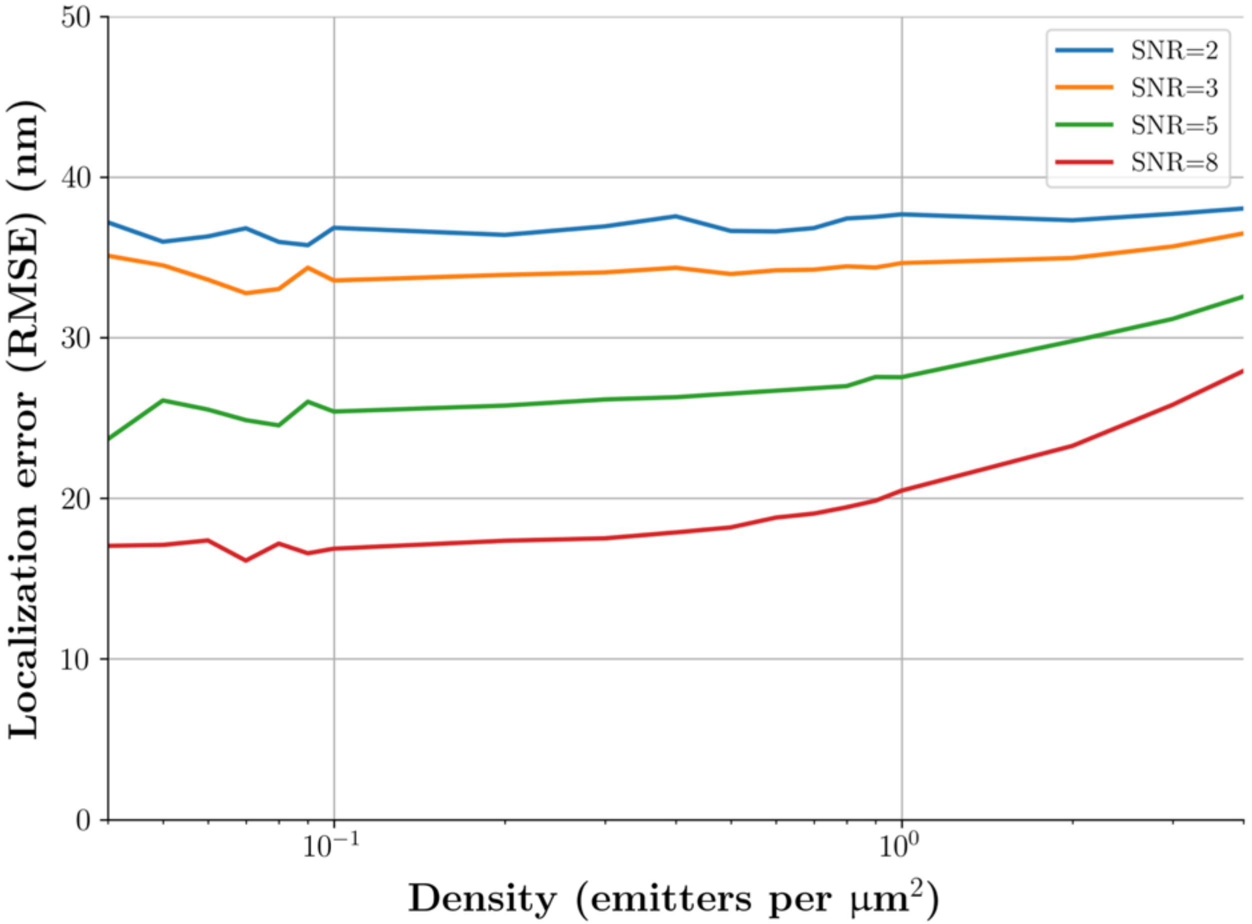


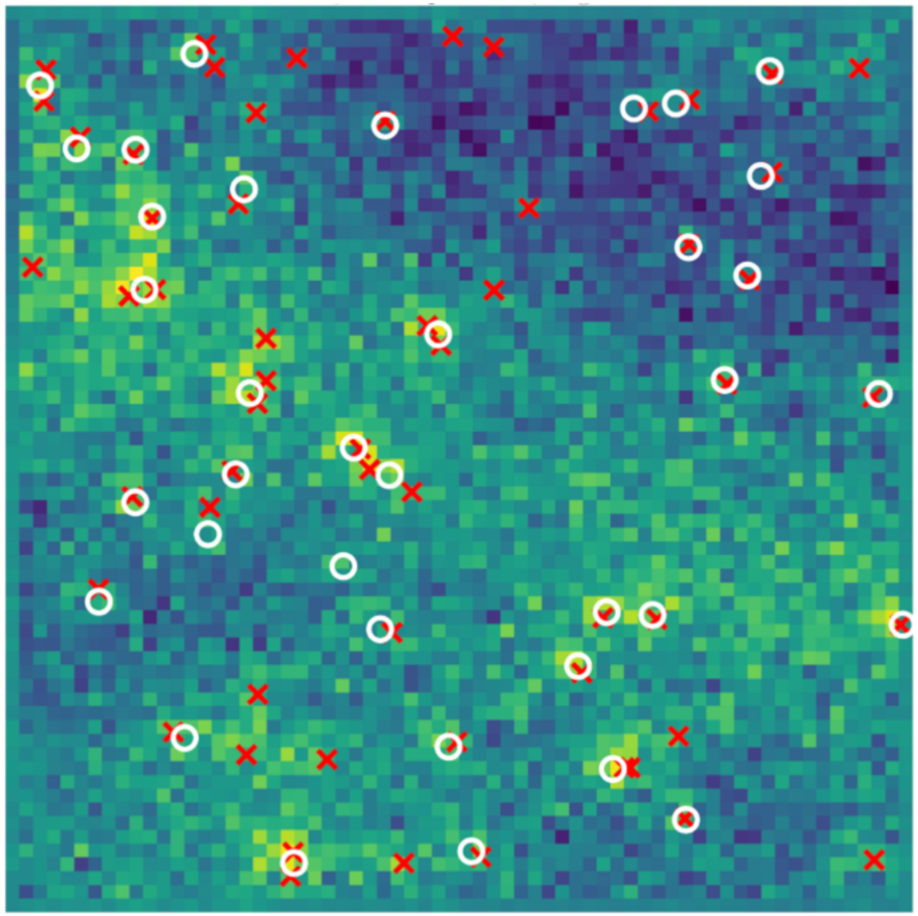
Jaccard Index = TP / (TP + FP + FN)



Density = 0.3 μm^{-2}
Avg. SNR of PSFs = 4.1

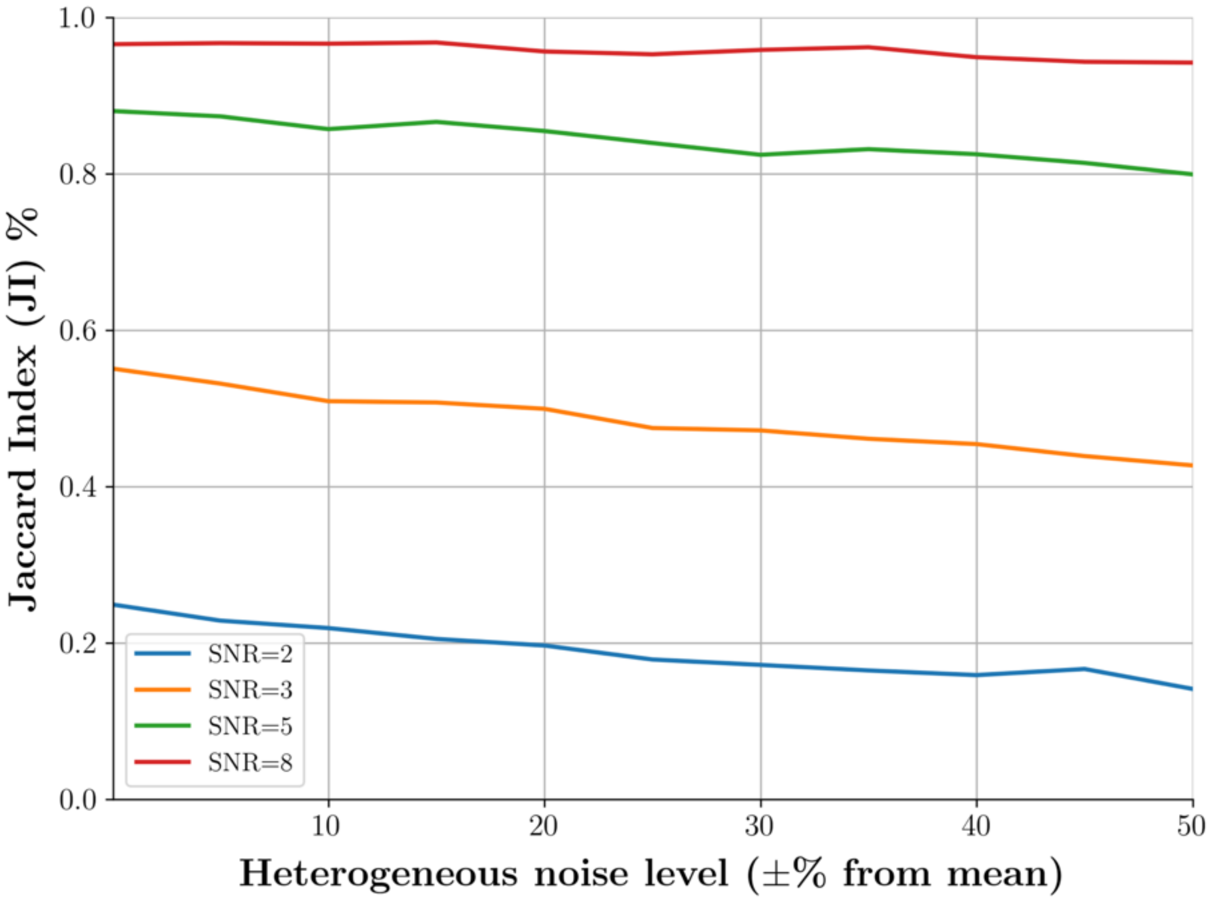
Localization error approaches limit of current methods

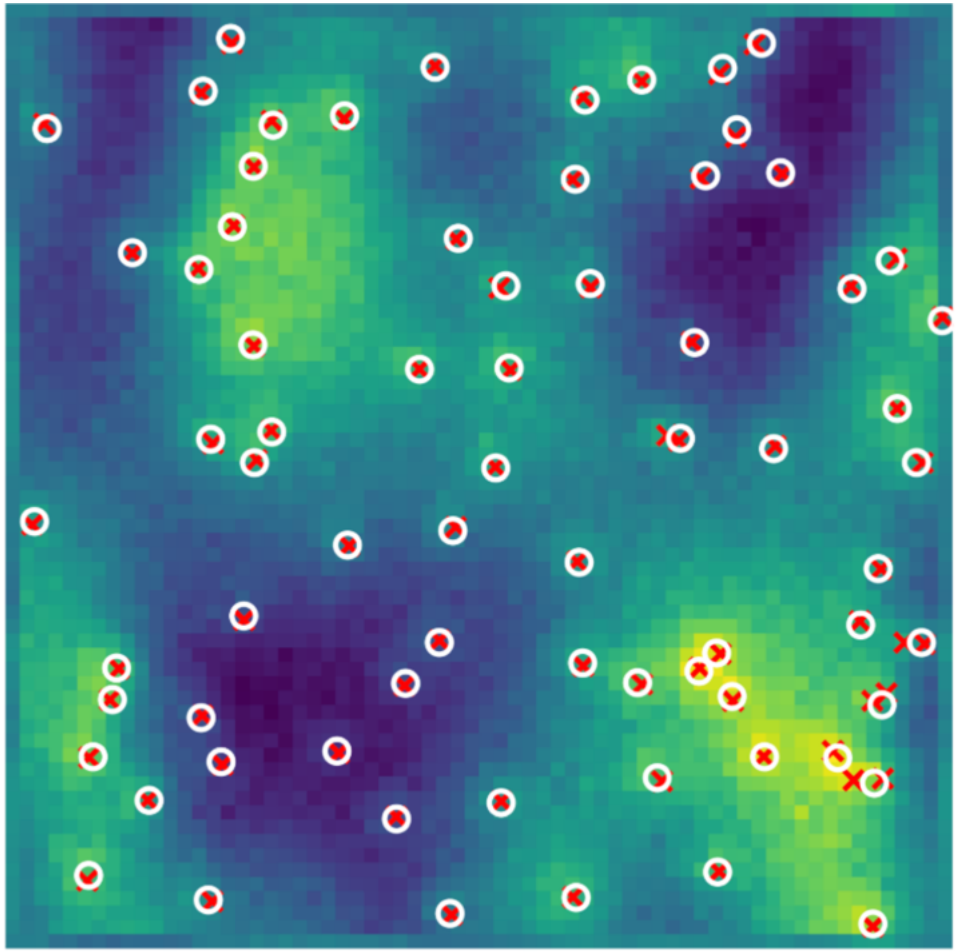




Density = 1.4 µm^{-2}
Avg. SNR of PSFs = 2.9

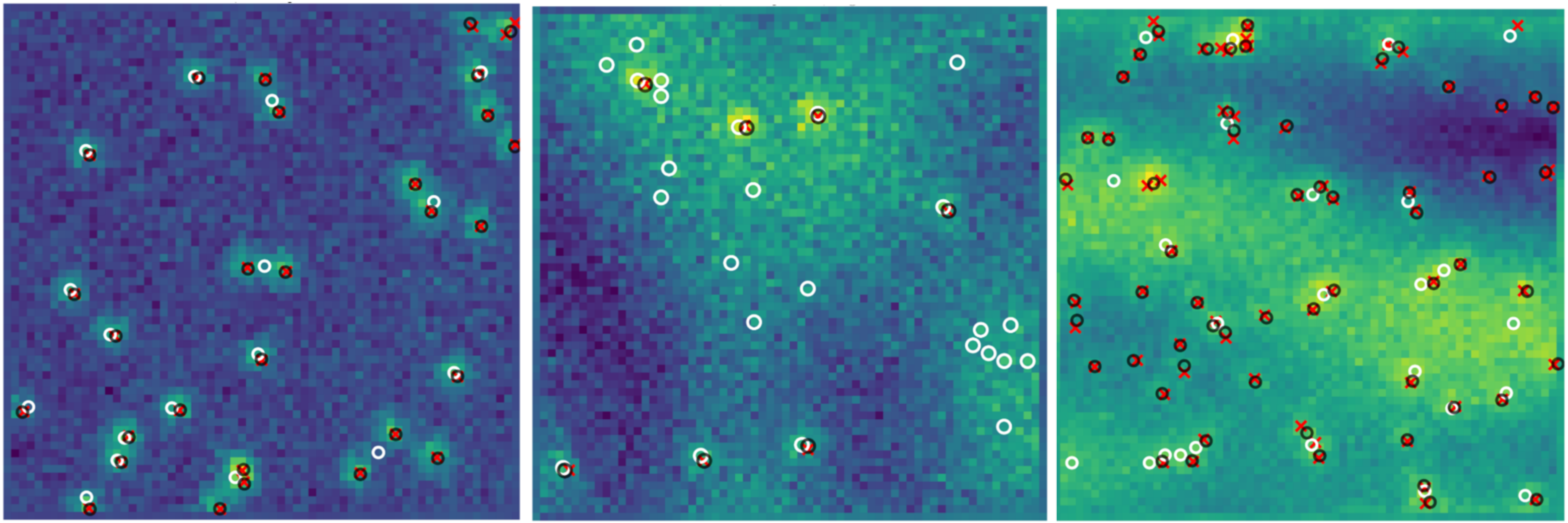
Accuracy decreases as heterogeneity increases





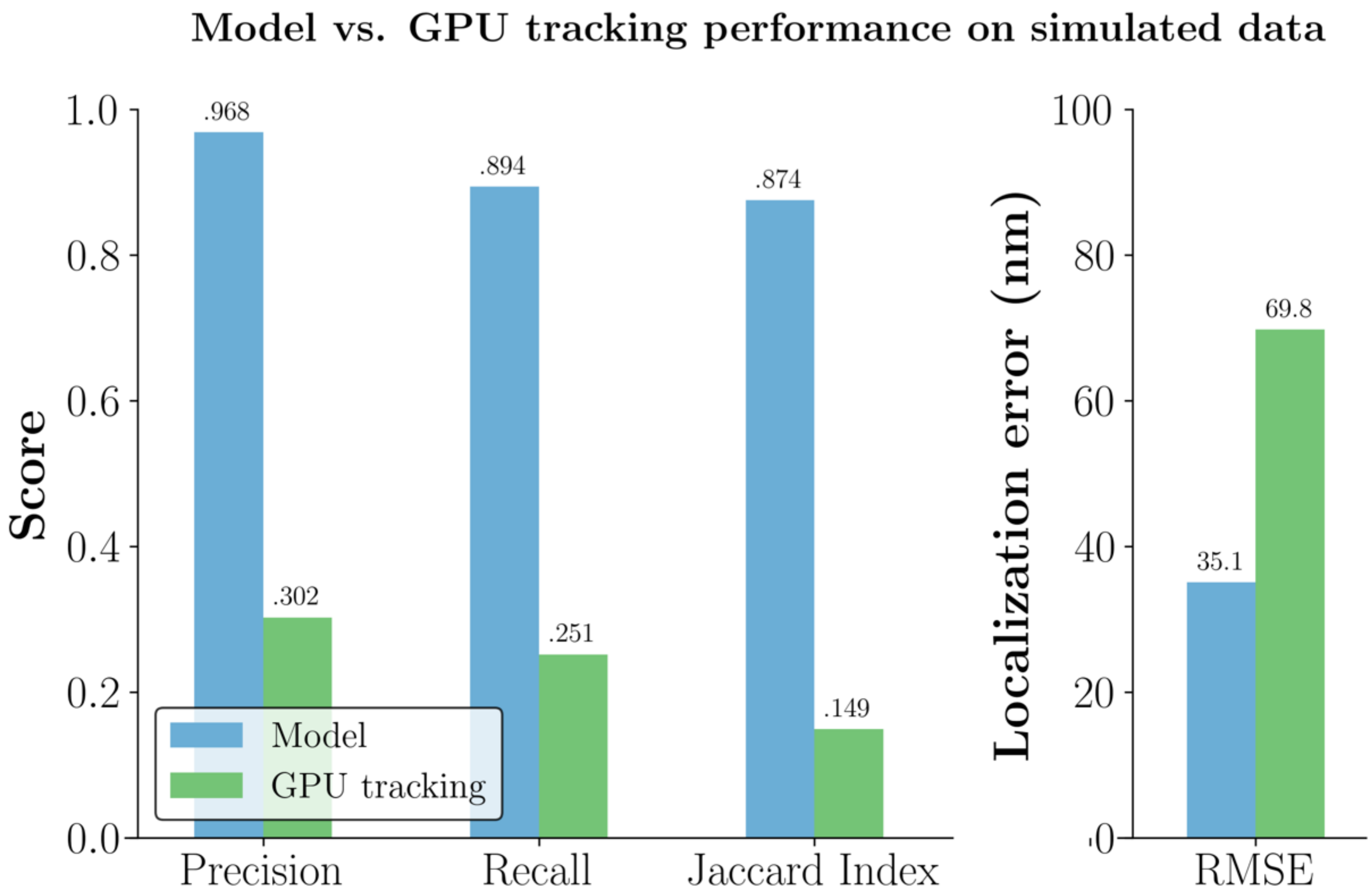
Density = 1.8 μm^{-2}
Avg. SNR of PSFs = 8.5

Model outperforms gpu-tracking, especially in heterogeneous background

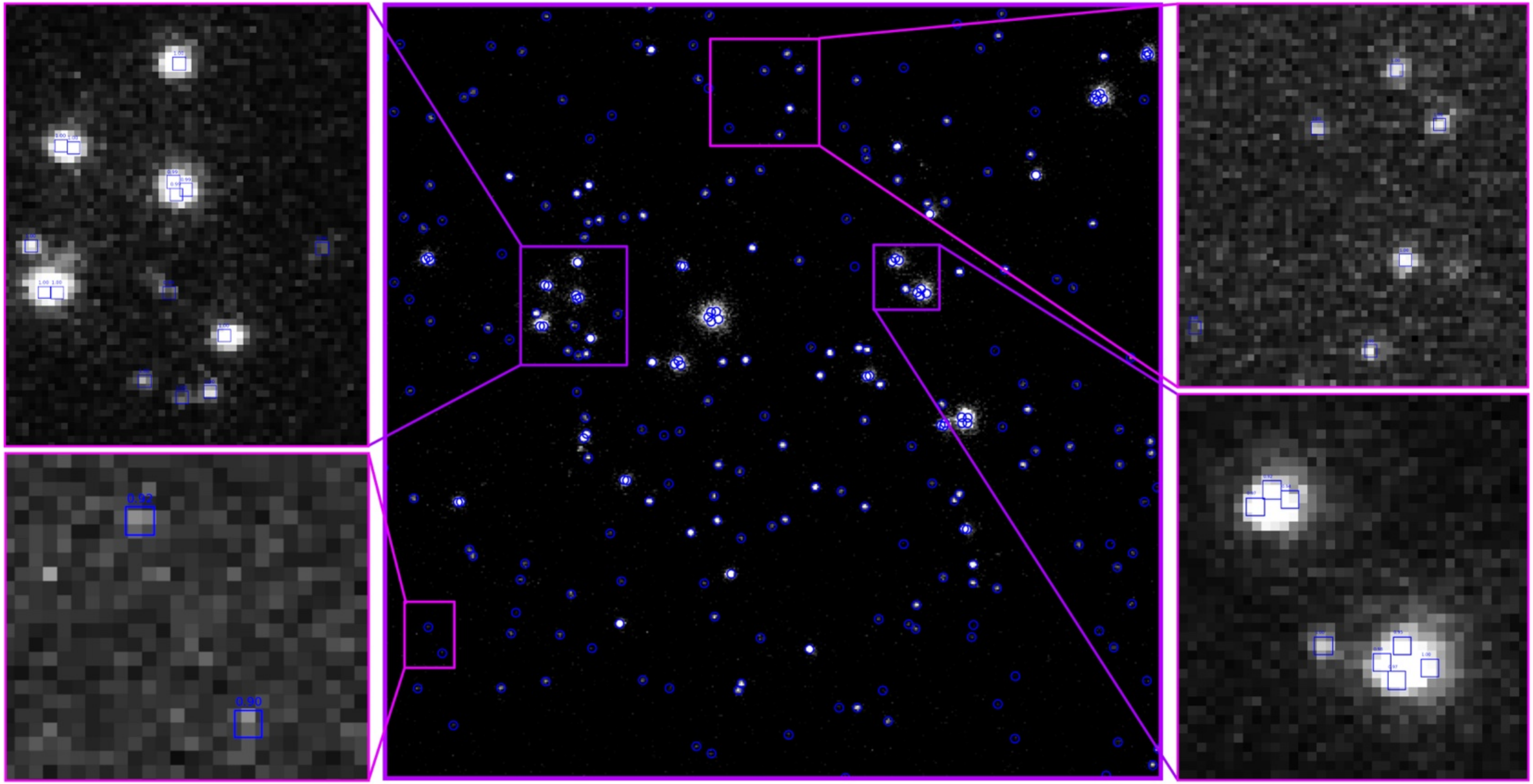


x ground truth
O gpu-tracking prediction
O model prediction

Comparing to conventional methods

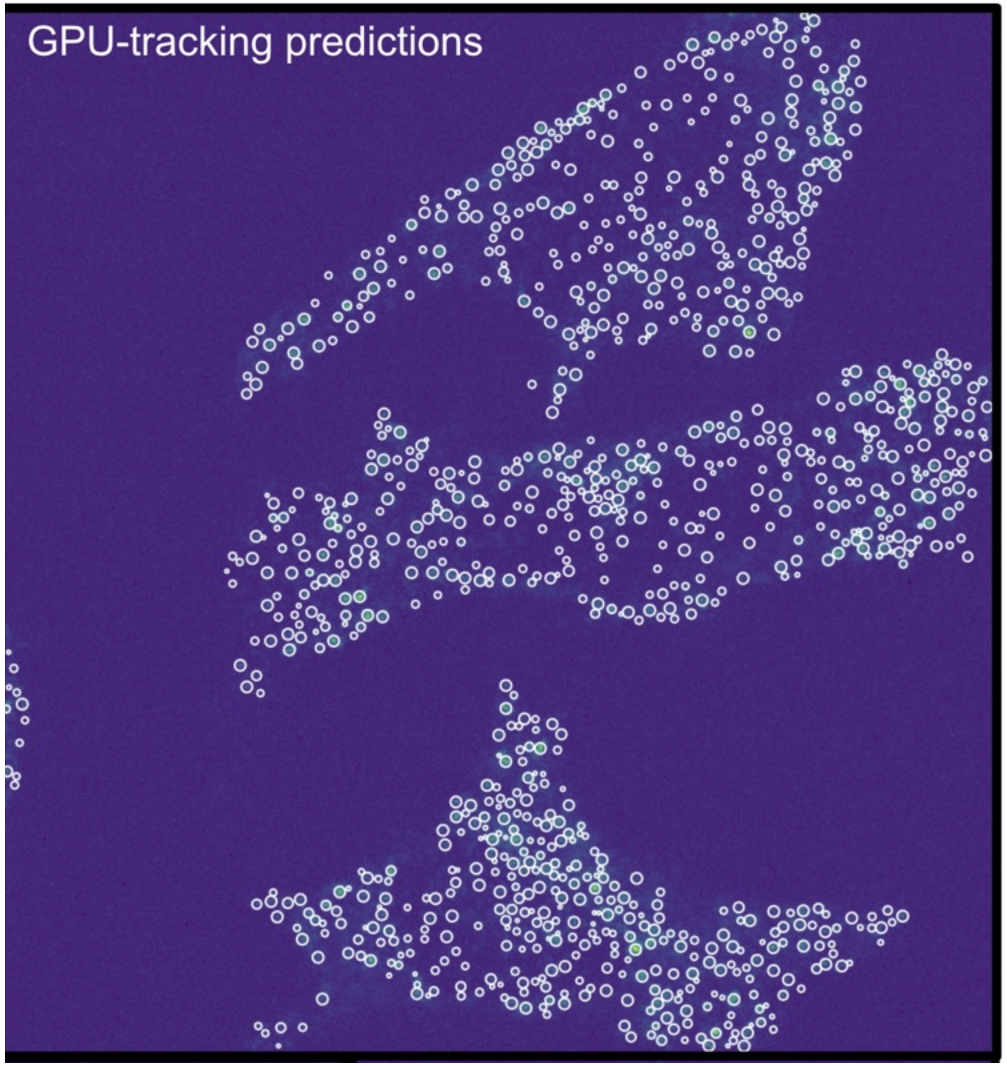


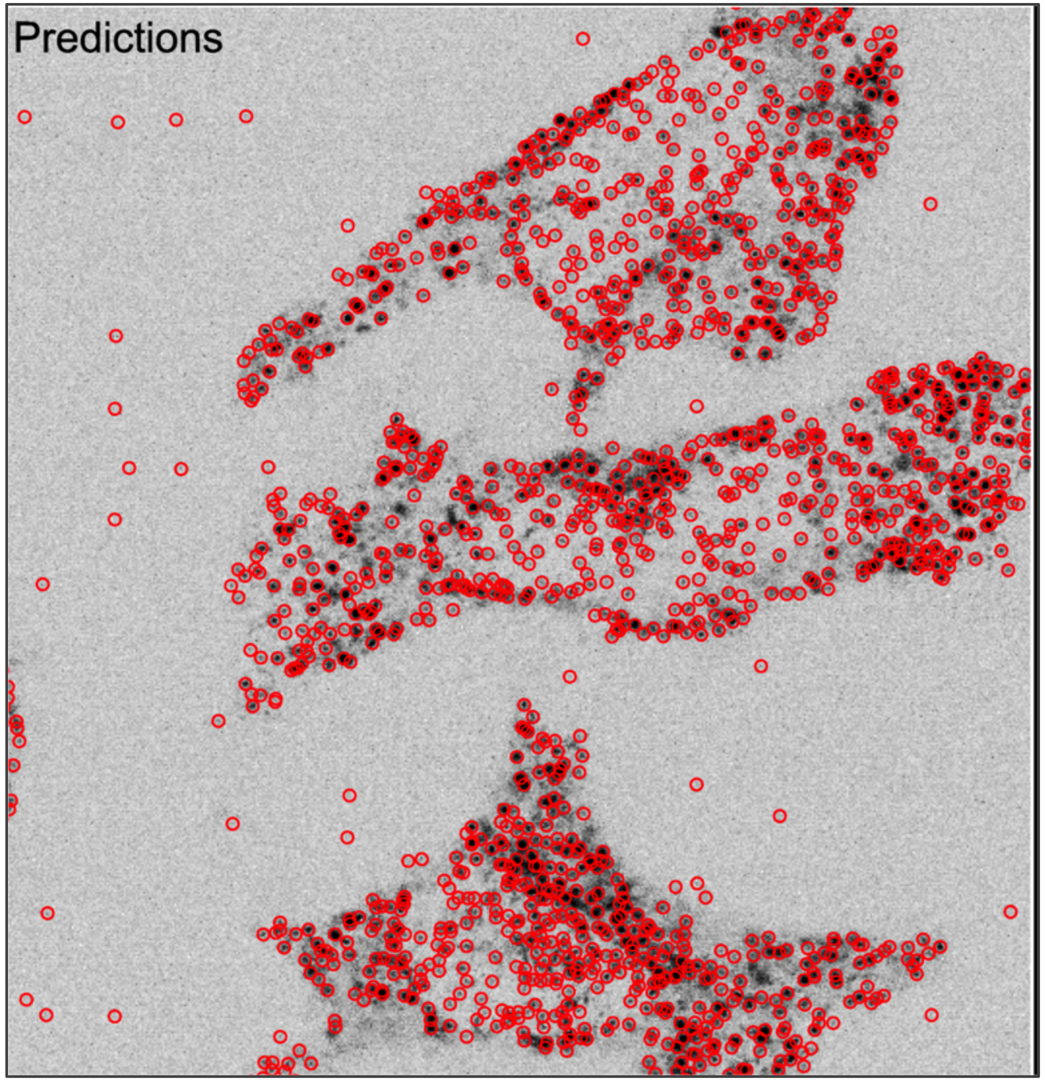
Results – experimental data



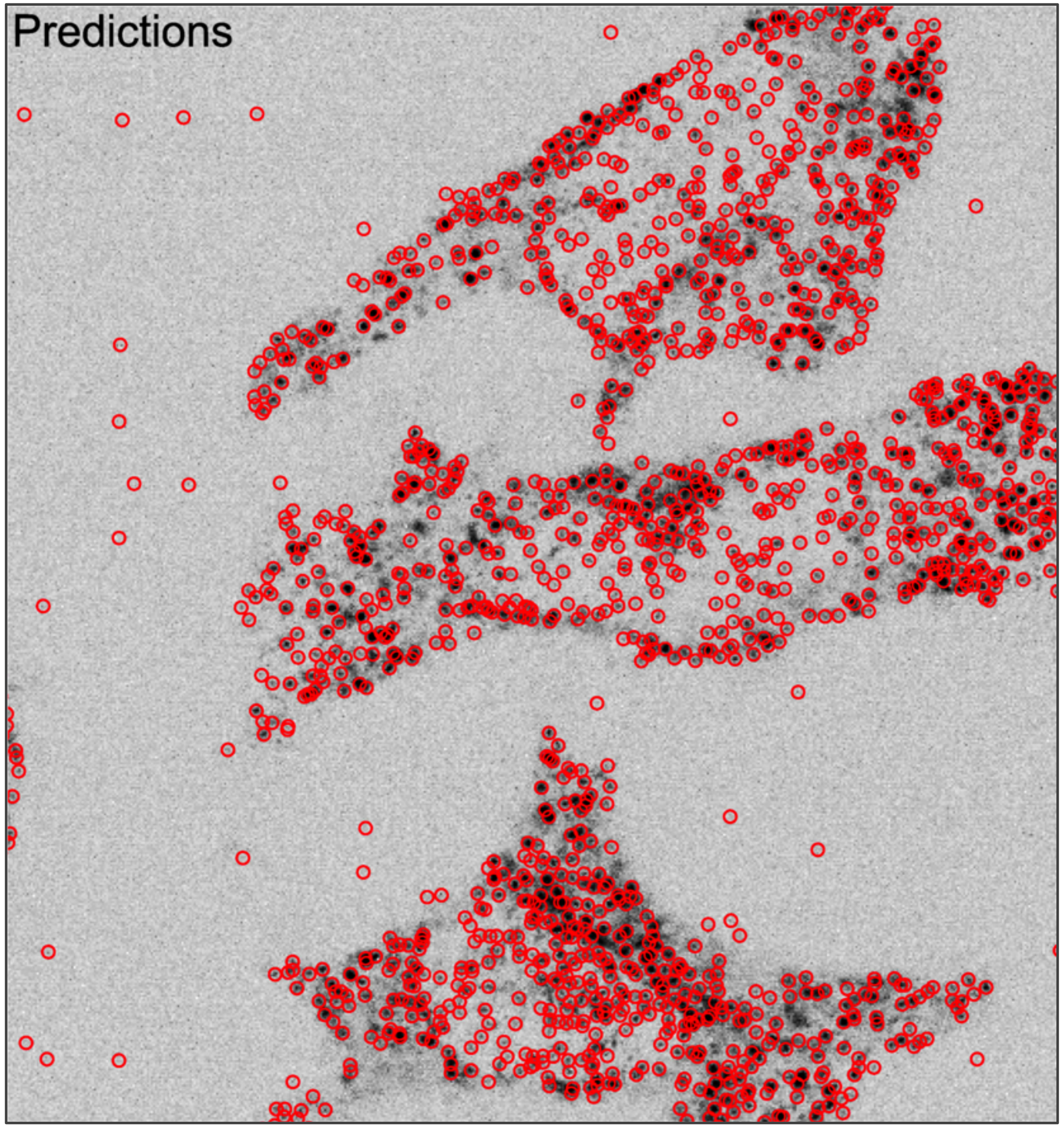
Collected by
Theodor Redvall

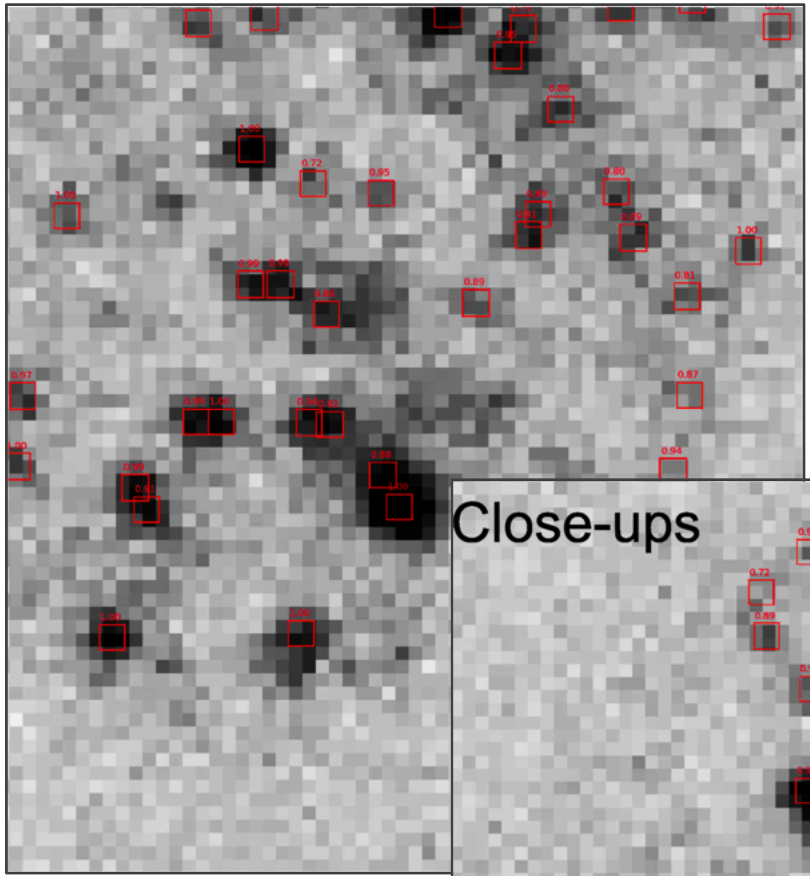
Qualitative comparison to gpu-tracking – experimental data

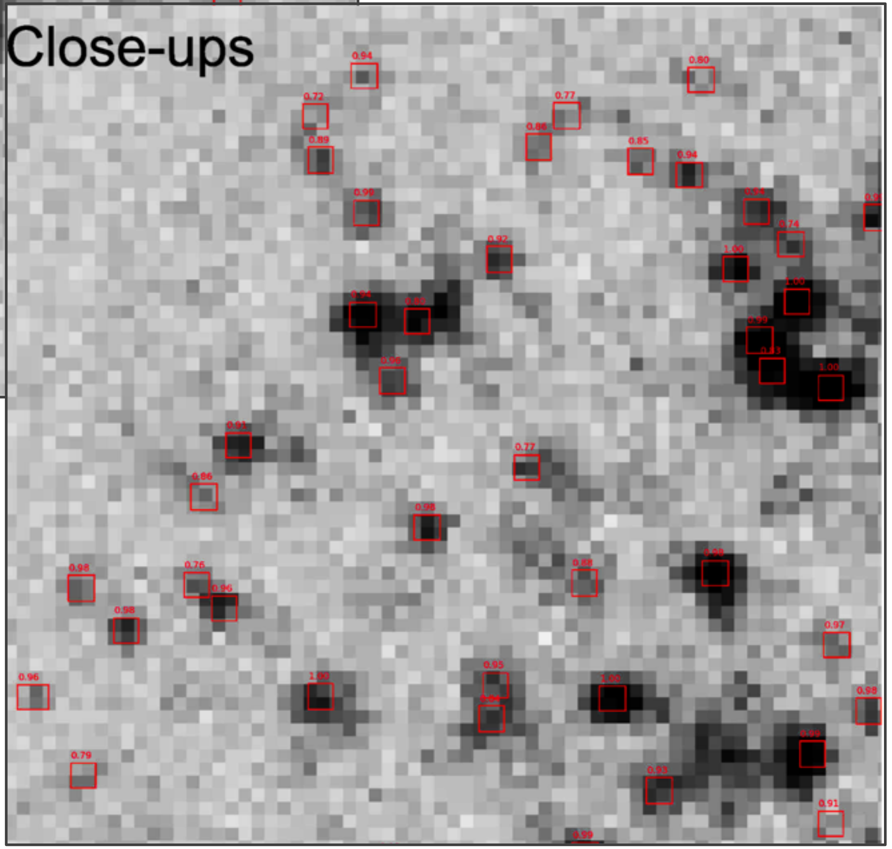




Results – experimental data







Conclusion: A Flexible, High-Accuracy PSF Localization Framework

- **No parameter tuning required**
  - Works out of the box - adapts to your data
- **High accuracy of detection**
  - 80-90% detection accuracy with $<$30nm localization error
- **Robust to variable imaging conditions**
  - Performs well in heterogeneous and noisy background environments
- **ROI-based deep-learning approach**
  - Ready for future non-PSF detection and task-specific predictions

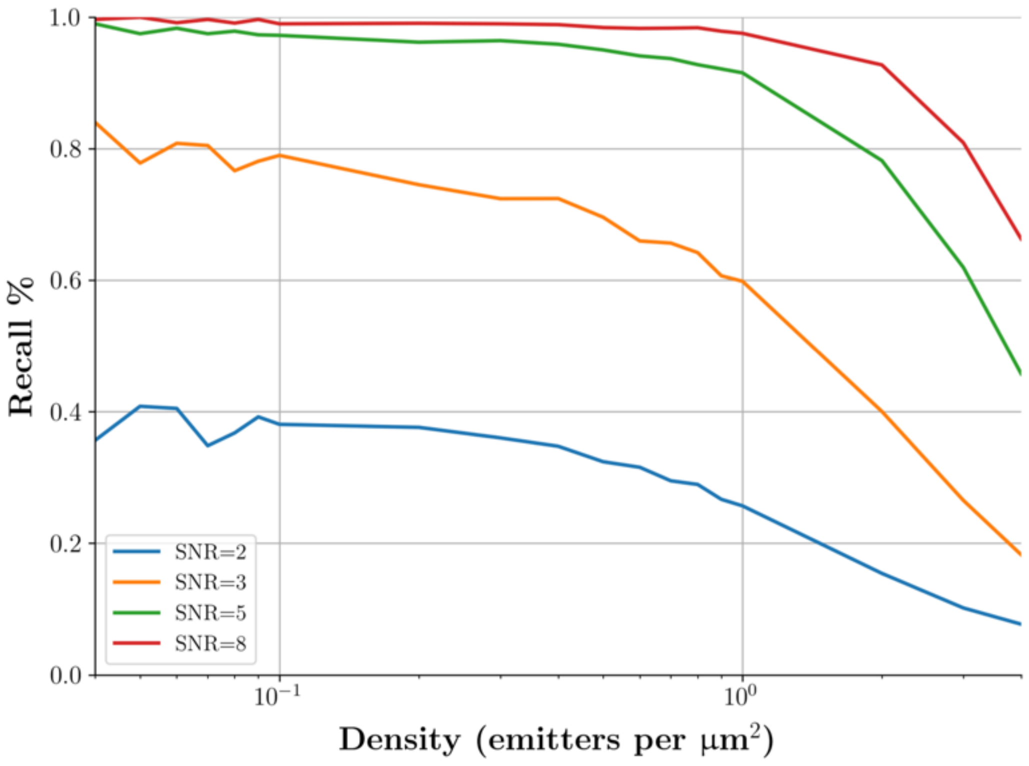
Future directions

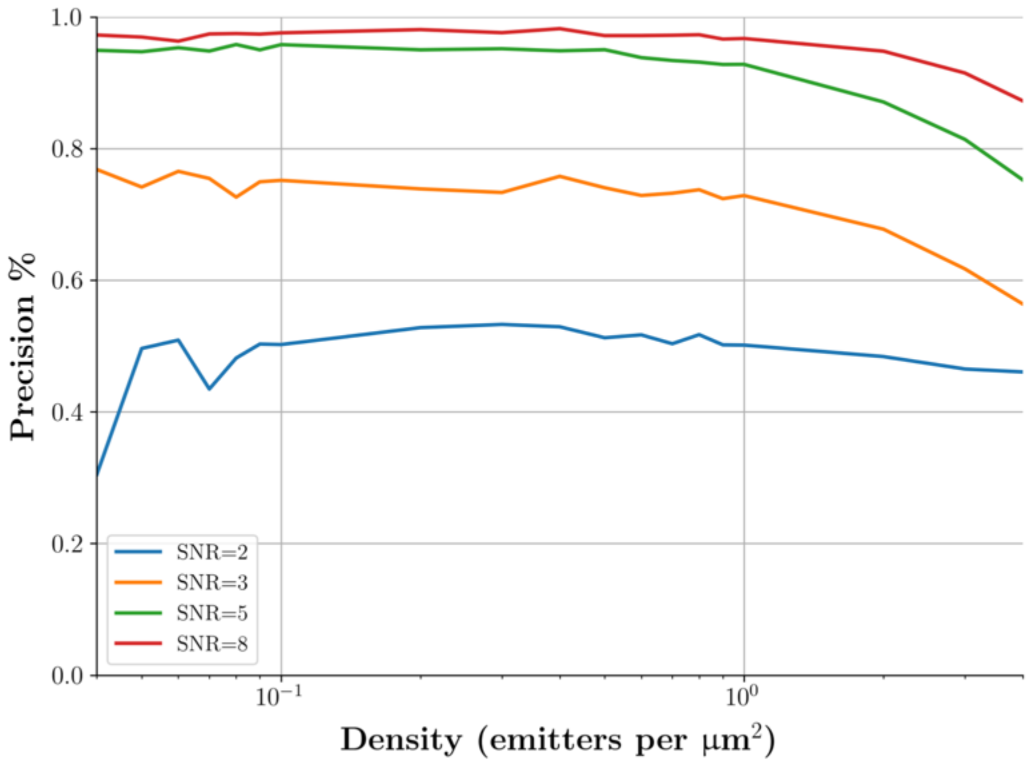
- Validate on third party datasets
- Test performance rigorously in edge cases
- Tune the Region Proposal Network

- Implement non-PSF detection and classification
- 3D images
- Temporal dimension

Thank you!

Detection accuracy increases with SNR and decreases with density of emitters

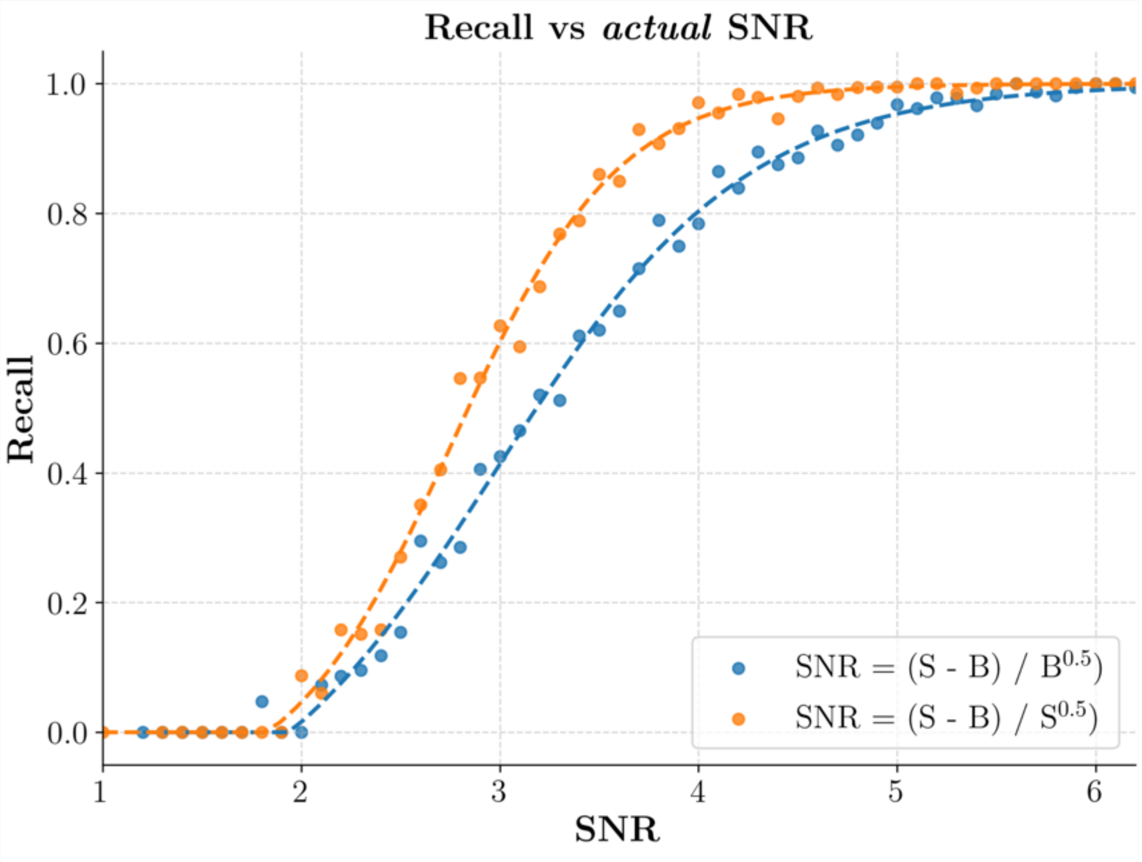


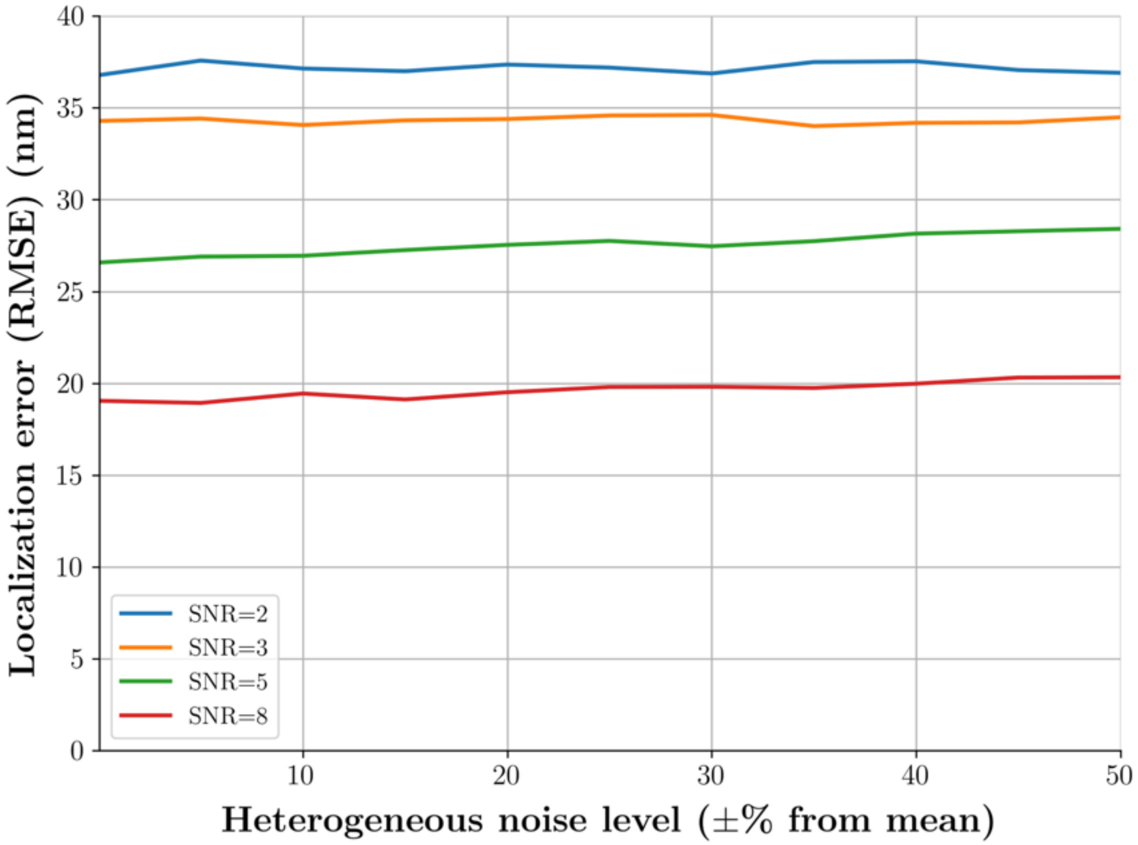


Recall = TP / (TP + FN)

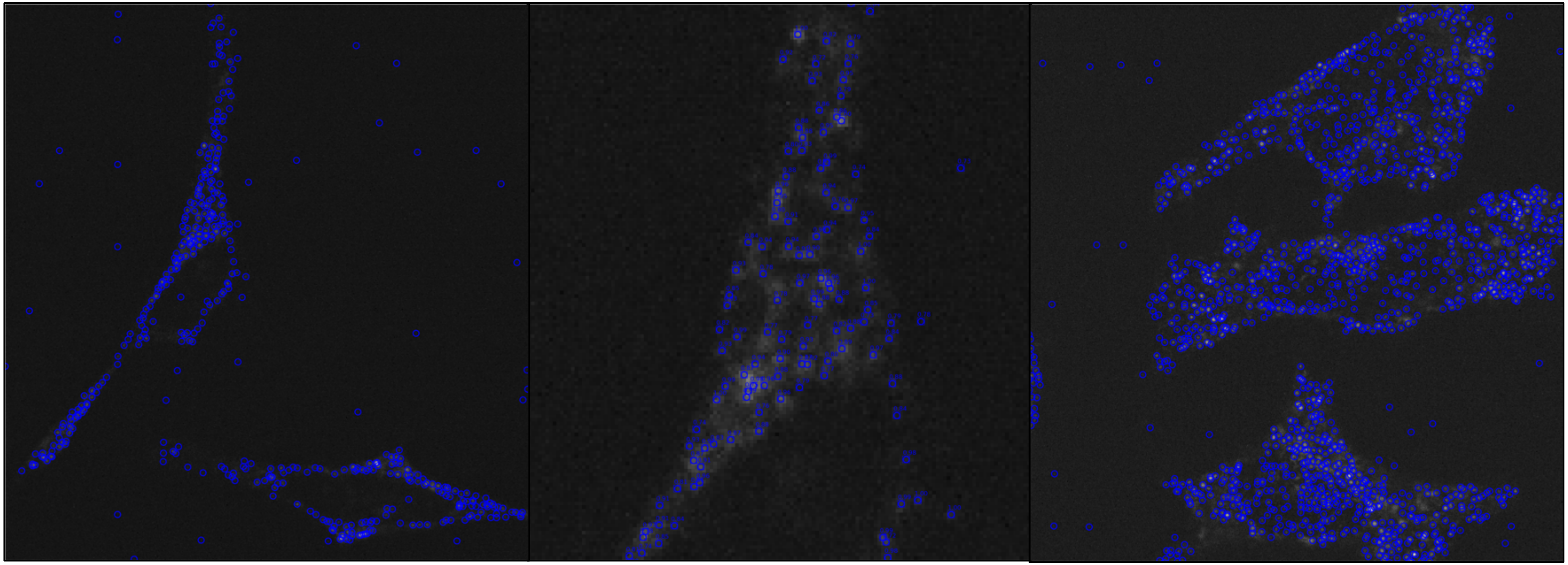
Precision = TP / (TP + FP)

Recall increases with signal-to-noise ratio. Localization accuracy decreases with background heterogeneity

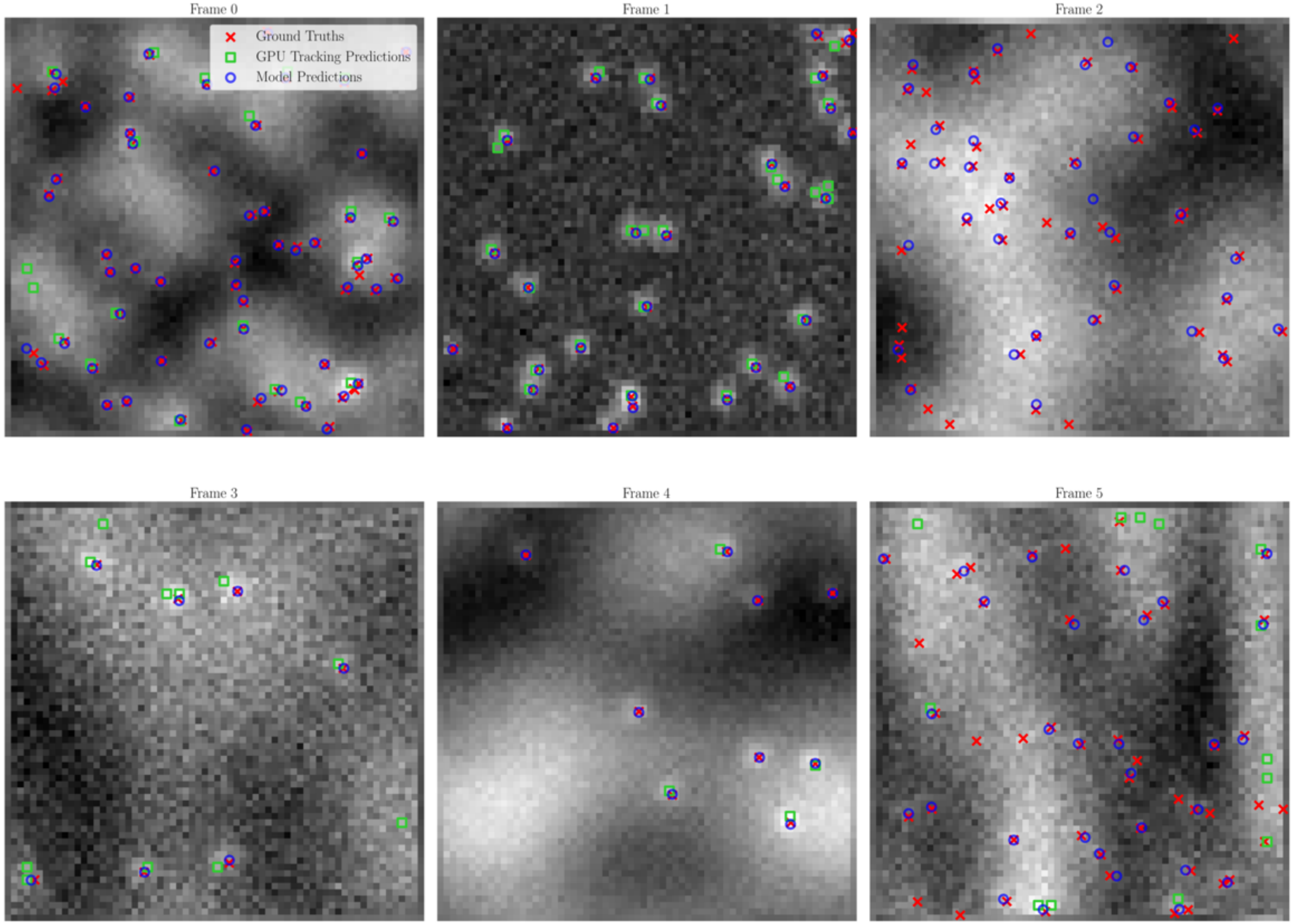




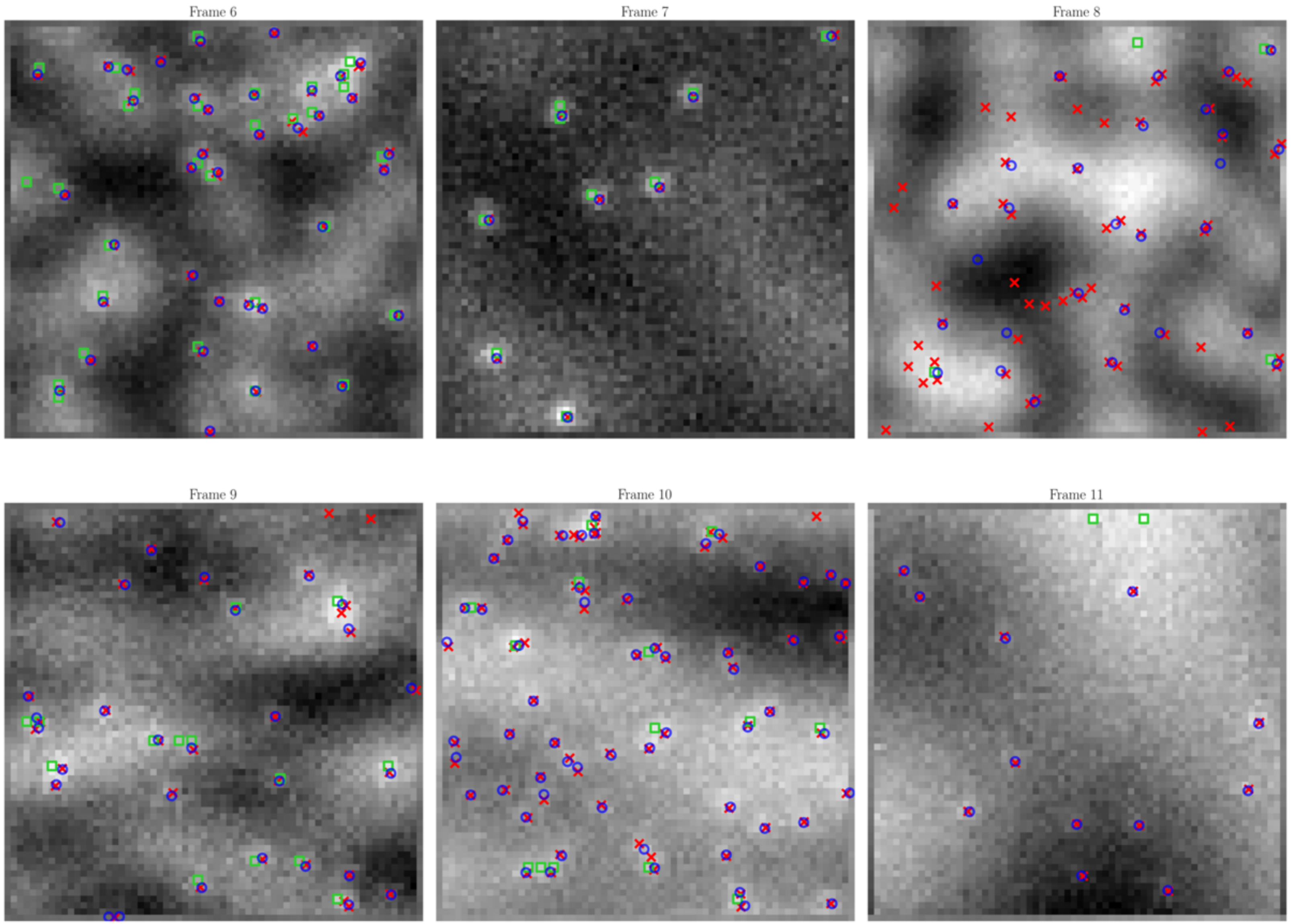
Model trained on images including zero emitters

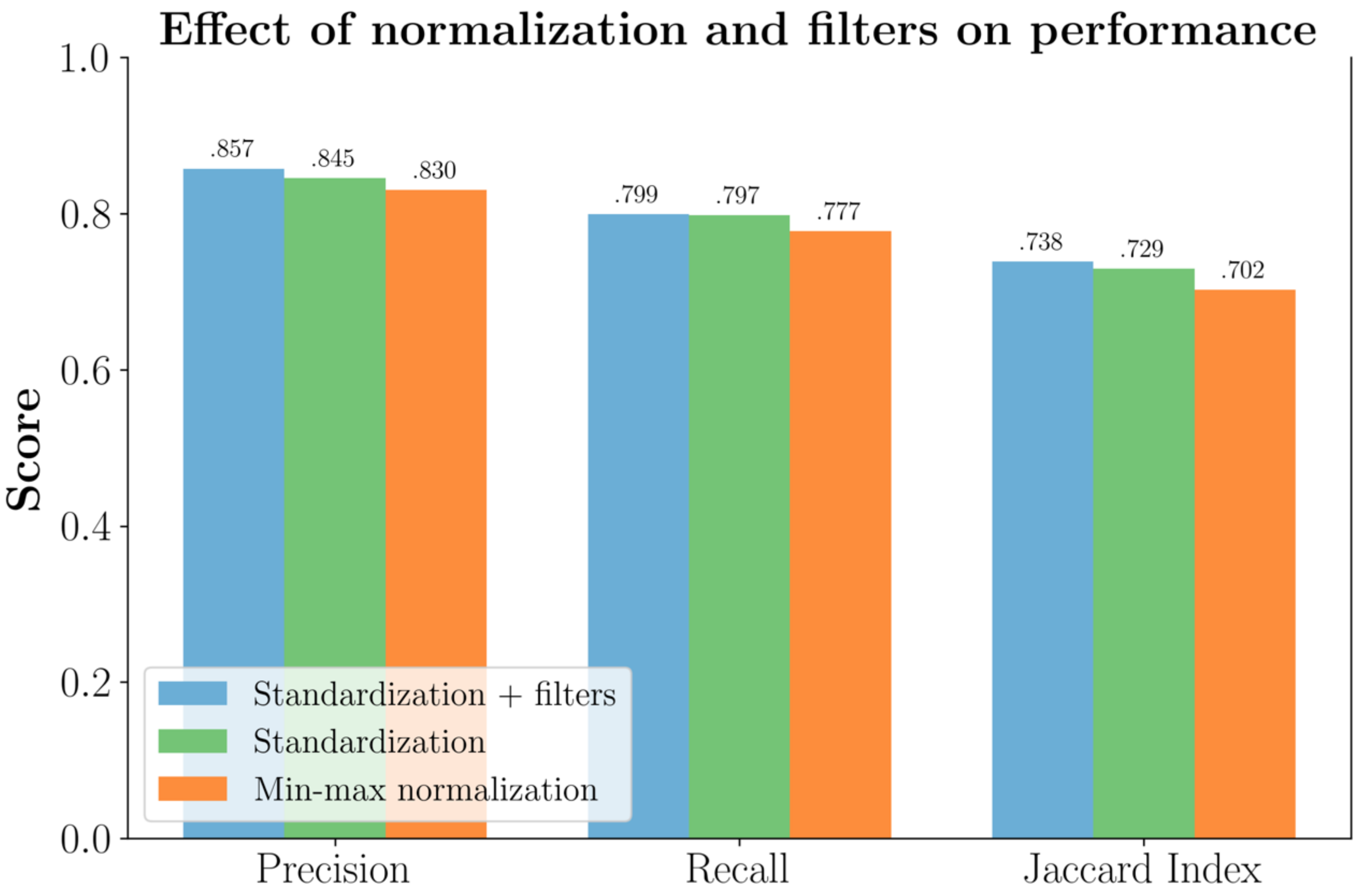


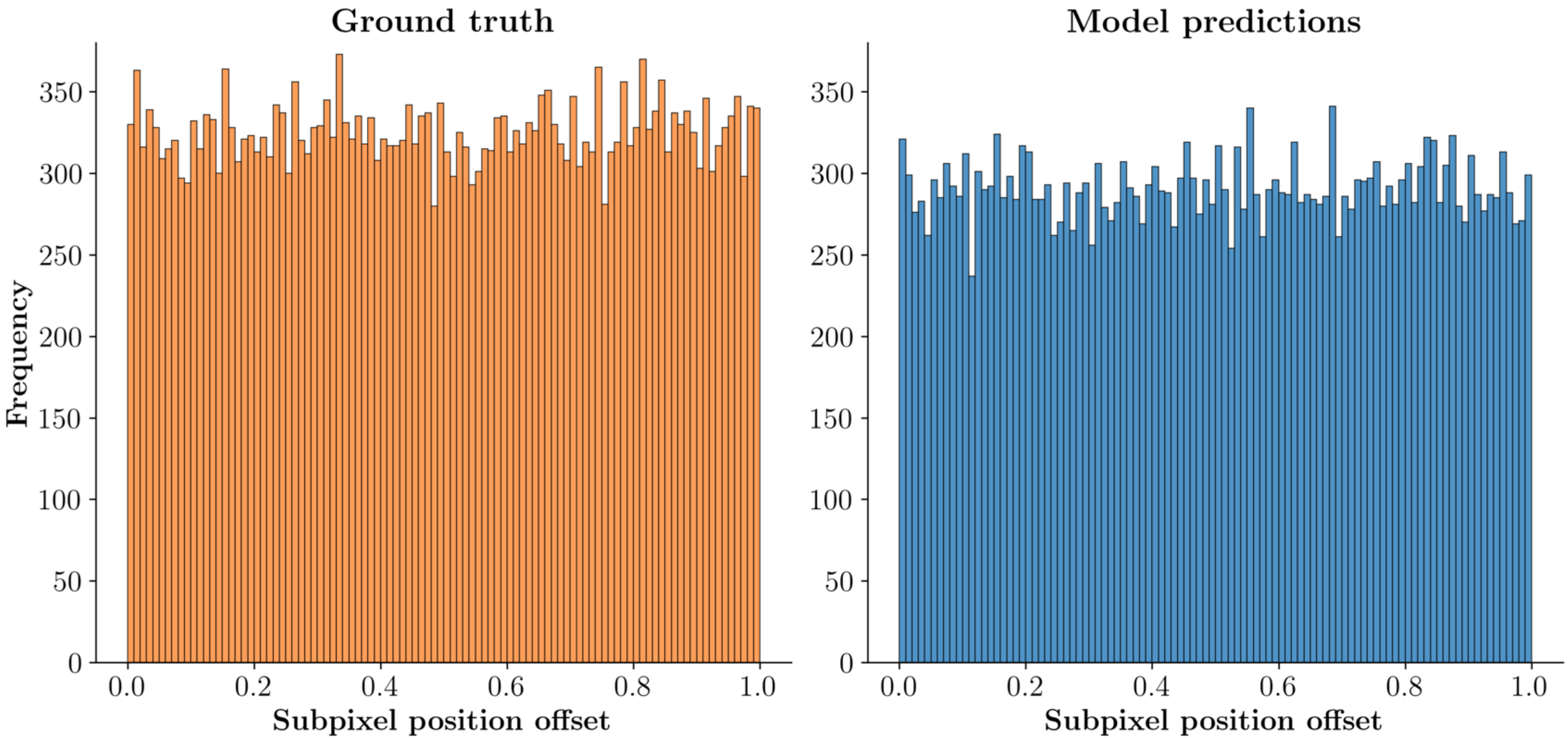
Appendix


Frame 0
Ground Truths
GPU Tracking Predictions
Model Predictions
Frame 1
Frame 2
Frame 3
Frame 4
Frame 5

Appendix

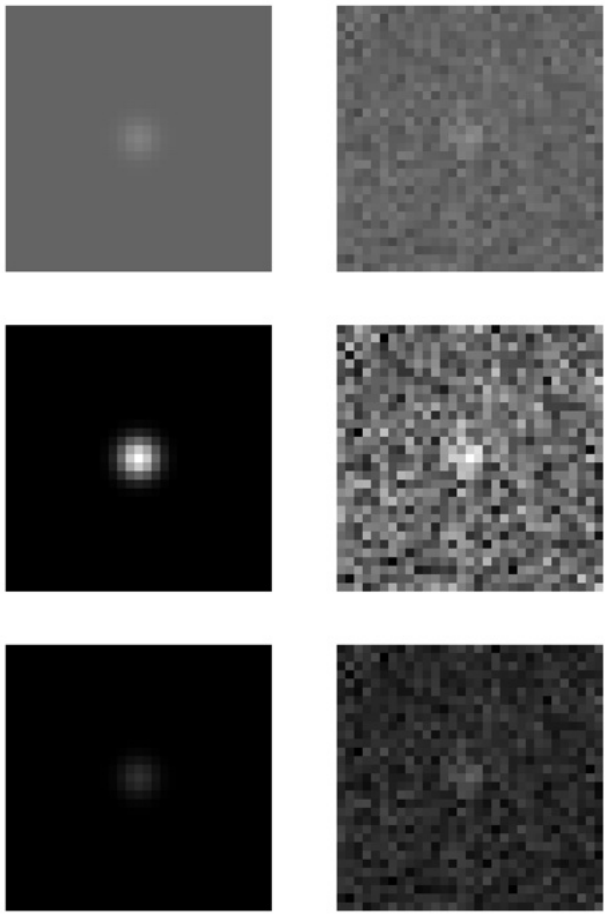



Effect of normalization and filters on performance
Score
1.0
0.8
0.6
0.4
0.2
0.0
.857
.845
.830
.799
.797
.777
.738
.729
.702
Standardization + filters
Standardization
Min-max normalization
Precision
Recall
Jaccard Index

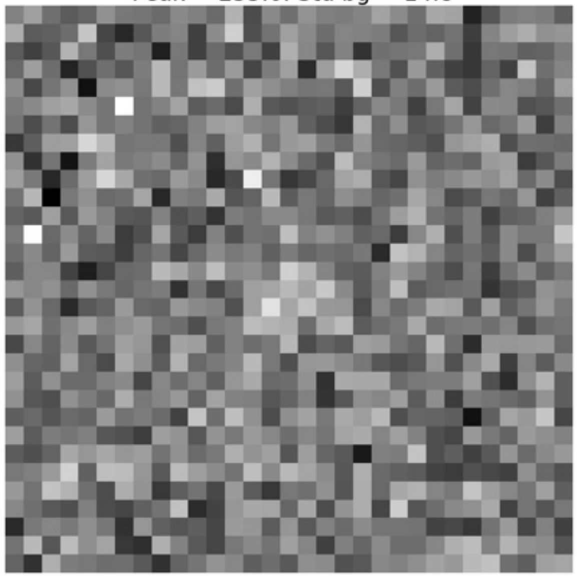

Ground truth
Model predictions
Frequency
0
50
100
150
200
250
300
350
0.0
0.2
0.4
0.6
0.8
1.0
Subpixel position offset
Subpixel position offset

SNR examples

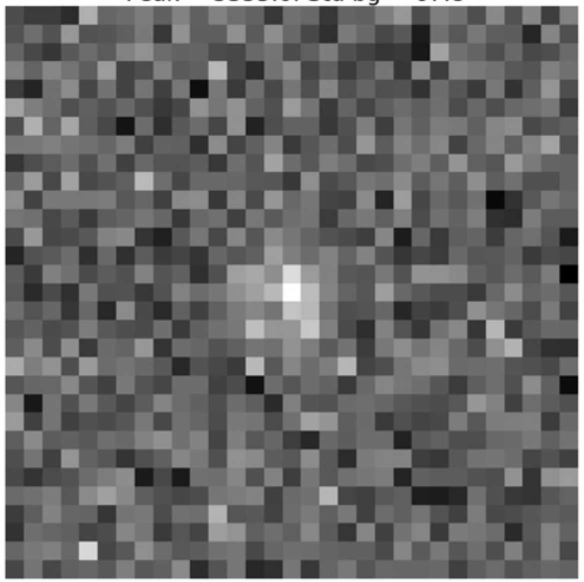
SNR_s = 3.6, SNR_bg = 4.3



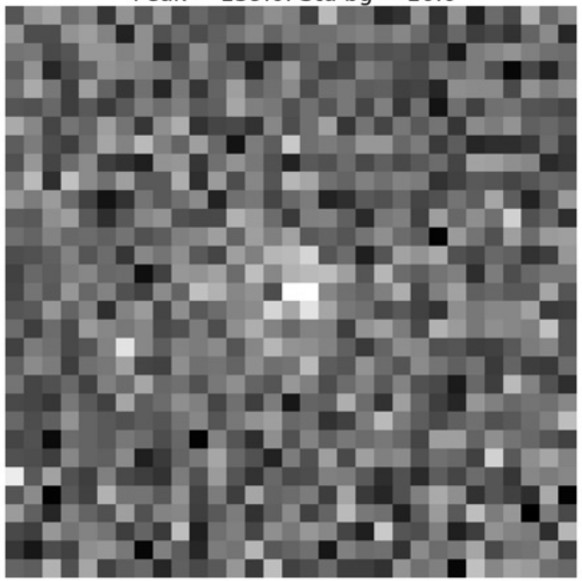
SNR_sig: 3.4, SNR_bg: 3.8. Mean bg = 201.3.
Peak = 255.0. Std bg = 14.8



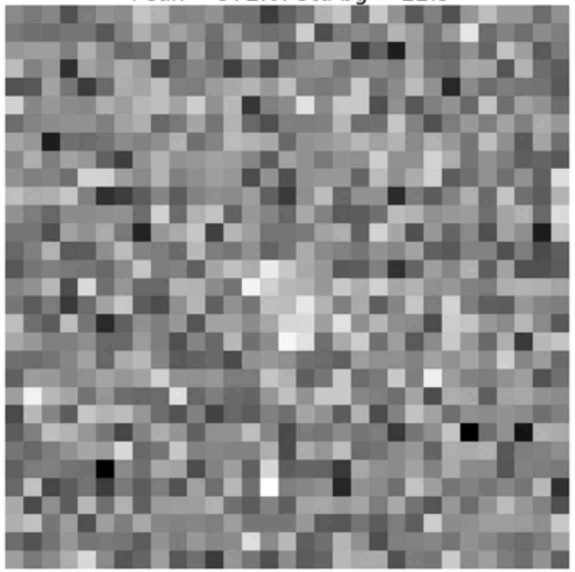
SNR_sig: 4.8, SNR_bg: 5.0. Mean bg = 5001.0.
Peak = 5355.0. Std bg = 67.8



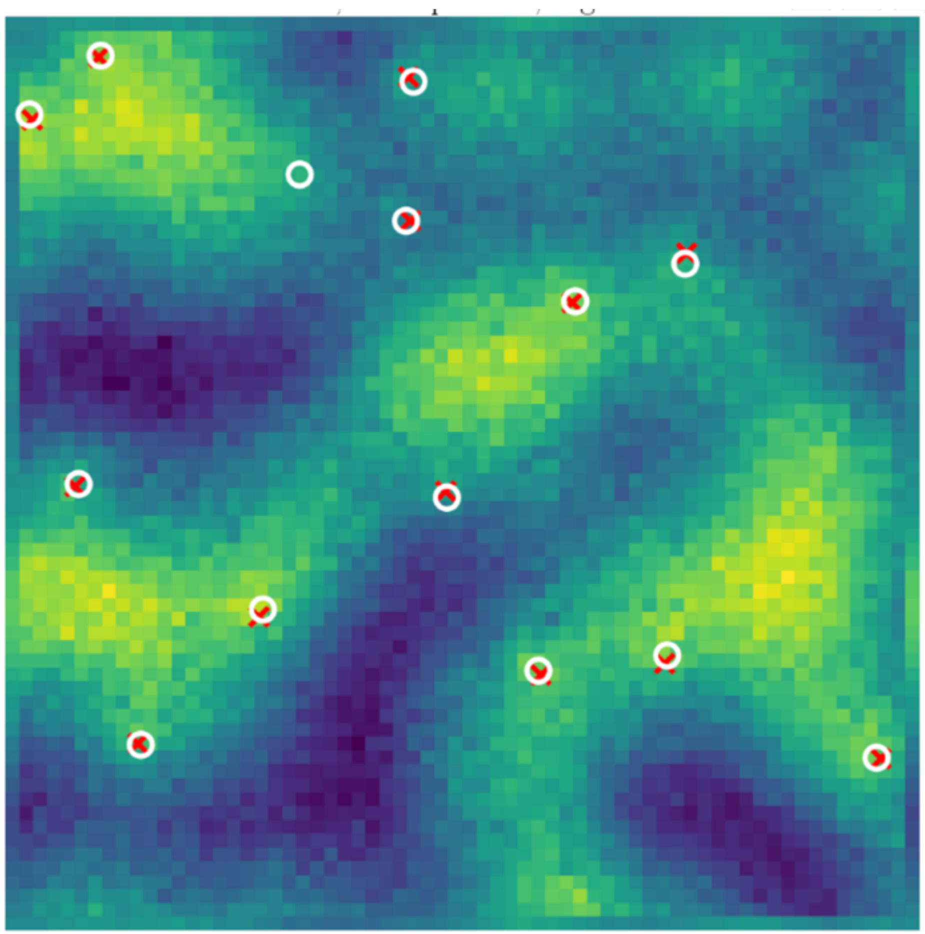
SNR_sig: 3.3, SNR_bg: 3.9. Mean bg = 100.0.
Peak = 139.0. Std bg = 10.0



SNR_sig: 3.0, SNR_bg: 3.2. Mean bg = 500.5.
Peak = 572.0. Std bg = 22.5



Test yourself



Qualitative comparison to gpu-tracking